Chapter 5

Venous Thromboprophylaxis

Two words best characterize the mortality and morbidity due to venous thromboembolism in the United States: substantial and unacceptable.

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To supplement Dr. Moser's statement, venous thromboembolism (i.e., deep venous thrombosis and pulmonary embolism) has four "alliterative" attributes: i.e., it is common, often covert, can be catastrophic, but is correctable. The risk of venous thromboembolism is almost universal in the patients who inhabit medical and surgical ICUs. Thrombus formation is most prominent in the major veins that drain the lower extremities (i.e., the femoral and iliac veins), and it often remains clinically silent until a portion (or all) of the thrombus breaks free and travels to the lungs to become a pulmonary embolus, which can be a life-threatening and even fatal event. Fortunately, there are a variety of preventive measures that can reduce or eliminate the risk of this life-endangering process.

Venous thromboembolism has been cited as the leading cause of preventable deaths in hospitalized patients, and preventive measures thus become "the number one priority for improving patient safety in hospitals" (1). This chapter addresses this mandate by describing the various methods used to prevent venous thromboembolism (i.e., *venous thromboprophylaxis*), and which methods work best in different clinical situations. Many of the recommendations in this chapter are taken from the clinical practice guidelines and reviews listed at the end of the chapter (2–11). (*Note:* The diagnosis and treatment of venous thromboembolism are described in Chapter 22.)

RISK FACTORS

There are a wide variety of risk factors for venous thromboembolism (VTE) in hospitalized patients, and these are organized in Table 5.1. The expression of these risk factors in different groups of patients is shown in Figure 5.1 (9).

Major Surgery

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Major surgery (i.e., surgery performed under general or spinal anesthesia that lasts longer than 45 minutes) is the most recognized cause of VTE in hospitalized patients, and is responsible for about one-third of deaths from VTE (4). All major surgical procedures create a heightened risk of VTE, which can persist for 6–12 weeks (i.e., after hospital discharge) (11). The risk is particularly high with cancer-related surgery (2,4,10), bariatric surgery (3), and orthopedic procedures involving the hip and knee (2,5).

The risk of VTE after major surgery is attributed to several factors, including thromboplastin release during the procedure (which produces a hypercoagulable state that lasts for days), inflammation (from prolonged procedures), vascular injury (especially in orthopedic and peripheral vascular procedures) and extended periods of bed rest in the perioperative period.

TABLE 5.1 Risk Factors for Venous Thromboembolism

Categories	Conditions
Surgery	Major surgery, especially hip and knee replacement, cancer-related surgery, and bariatric surgery.
Trauma	Traumatic brain injury, fractures of the spine, pelvis, hip, or lower extremities, extensive tissue and burn injuries.
Malignancy	Solid tumors (especially pancreatic and gastric CA), hematologic malignancies, chemotherapy.
Acute medical Illness	Acute stroke with lower extremity weakness, heart failure with leg edema, COVID-19, sepsis, limited mobility.
ICU-Related	Mechanical ventilation, neuromuscular paralysis, central venous catheters, circulatory shock, vasopressors, platelet transfusions, prolonged immobility.
Patient-Specific	Prior VTE, family history of VTE, genetic risk factors, age >70, obesity, pregnancy or postpartum.

Major Trauma

The risk of VTE is highest following major or multisystem trauma, where the reported incidence of DVT is 58% without prophylaxis (6). The trauma conditions with the highest risk of VTE are brain and spinal cord injuries, spinal fractures, and fractures of the pelvis, hip, and long bones of the legs. There are several factors that predispose to VTE in trauma, including thromboplastin release and inflammation from extensive tissue injury, vascular injury, long bone fractures (which release thrombogenic lipids into the circulation), and prolonged periods of immobility from lower extremity injuries.

Acute Medical Illness

Hospitalization for acute medical illness is associated with an eightfold increase in the risk of VTE (8). Although VTE is less common in medical patients than in surgical patients or trauma victims (see Figure 5.1), medical patients account for a majority (50% to 75%) of VTE events in hospitalized patients (8). The conditions that create the highest risk of VTE are neoplastic diseases, acute stroke with weakness of the extremities, and right-sided heart failure with leg edema. (*Note:* COVID-19 is associated with a high risk of VTE, but this illness seems to have

subsided.)

ICU Patients

As mentioned earlier, a majority of ICU patients have one or more of the risk factors for VTE listed in Table 5.1. There are also ICU-specific factors that add further to the risk of VTE, which include prolonged mechanical ventilation (>48 hrs.), systemic inflammatory conditions (e.g., sepsis), circulatory shock, vasopressor infusions, platelet transfusions, and the presence of central venous catheters or peripherally inserted central catheters (PICCs).

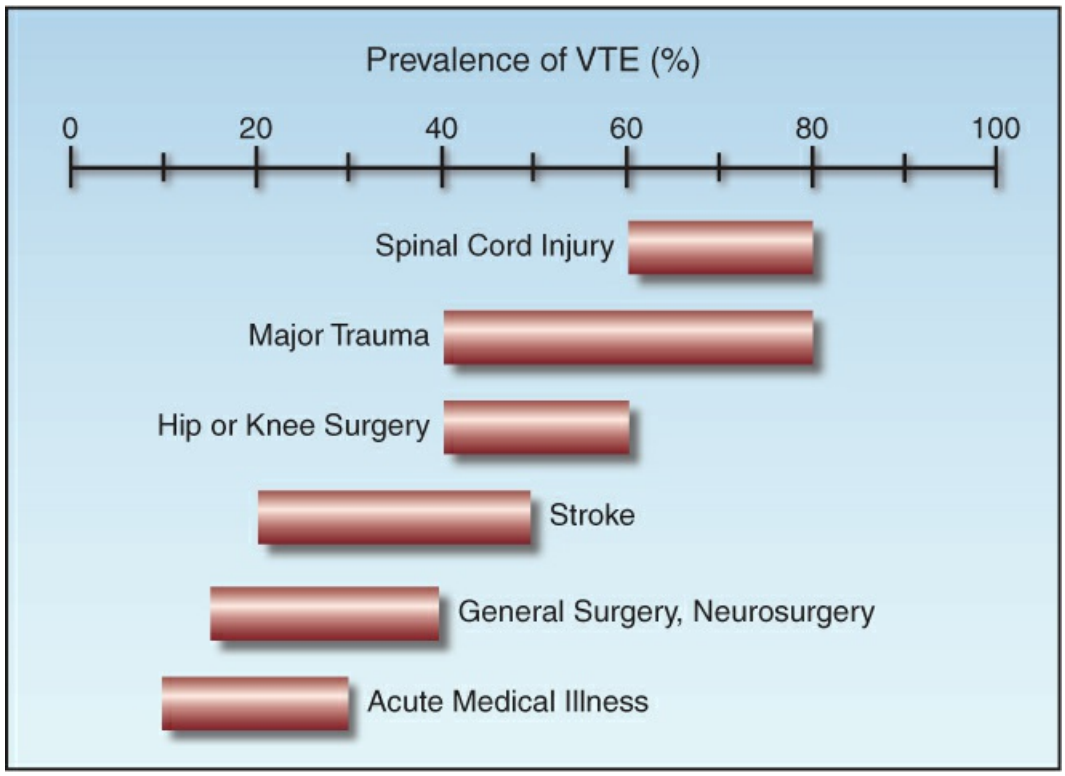


FIGURE 5.1 Risk of venous thromboembolism (VTE) without thromboprophylaxis, in different groups of patients. Includes both symptomatic and asymptomatic cases. From Reference 9.

Patient-Specific Factors

Patient–specific risk factors for VTE include advanced age, obesity, pregnancy, personal or family history of VTE, and genetic predisposition to thrombosis (e.g., Factor V deficiency, antithrombin III deficiency). The factors that confer the highest risk of VTE include a prior history of VTE, and the genetic risk factors; e.g., inherited antithrombin III deficiency creates a 14-fold increase in the risk of VTE (11).

PARENTERAL PROPHYLACTIC REGIMENS

A number of anticoagulant regimens have proven effective in reducing the incidence of VTE in hospitalized patients (2–8), but the following considerations are needed before starting an

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anticoagulant regimen:

. Absolute contraindications include intracranial hemorrhage and "clinically important" bleeding, which is defined as blood loss associated with a decrease in systolic blood pressure ≥20 mm Hg, a decrease in hemoglobin ≥2 g/dL, or the need to transfuse at least 2 units of packed red blood cells.

. Relative contraindications include traumatic head and spinal injury, a history of recurrent GI bleeding, and a severe coagulopathy (e.g., platelets <50,000).

. Menstrual bleeding is not a contraindication to anticoagulant prophylaxis.

Anticoagulant prophylaxis is typically achieved by subcutaneous injection, using the drug regimens summarized in Table 5.2.

TABLE 5.2 Thromboprophylaxis with Parenteral Agents

Agent	Prophylactic Dosing Regimen
Unfractionated Heparin	Standard: 5,000 Units SC every 8–12 hrs. Obesity: 7,500 units every 8 hrs for BMI ≥ 40 kg/m^2. Renal: No dose adjustment.
Enoxaparin (Lovenox)	Standard: 40 mg SC once daily, or 30 mg SC every 12 hrs. Obesity: 40 mg SC every 12 hrs for BMI ≥ 40 kg/m^2. Renal: If Cr CL < 30 mL/min, reduce dose to 30 mg SC once daily, or use UFH.
Dalteparin (Fragmin)	Standard: 2,500–5,000 units SC once daily. Obesity: Not adequately studied. Renal: No dose adjustment.
Fondaparinux (Arixtra)	Standard: 2.5 mg SC once daily. Obesity: No recommendations. Renal: If Cr CL = 35–50 mL/min, reduce dose to 1.5 mg SC once daily. Do not use if Cr CL < 30 mL/min.

From References 14, 15, 17–20, 24, 25. SC = subcutaneous, Cr CL = creatinine clearance, UFH = unfractionated heparin.

Unfractionated Heparin (UFH)

Heparin is a naturally occurring mucopolysaccharide that is made by mast cells in a wide variety of species, including humans. (Porcine intestines and bovine lungs have been the traditional sources of heparin.) Native or "unfractionated" heparin (UFH) is a heterogeneous collection of molecules that vary in size (molecular weight) from 3,000 to 30,000 Daltons (12), and this size variation has an important influence on anticoagulation produced by heparin, as described next.

Mechanisms

Heparin exerts its anticoagulant effect in two ways. First, it binds to antithrombin III (AT) in circulating blood, and this activates the AT and leads to inhibition of activated factor X (Xa), as shown in Figure 5.2. Since factor Xa normally converts prothrombin to thrombin, the inhibition

of factor Xa ("anti-Xa" effect) inhibits the generation of thrombin. In addition to this action, the heparin molecule forms electrostatic bonds with thrombin, and this inhibits the ability of thrombin (factor IIa) to convert fibrinogen to fibrin. This "anti-IIa" effect is size-dependent (i.e., dependent on the length of the polysaccharide chain in the heparin molecule), and this explains why the size variation in UFH preparations produces a variable anticoagulant effect. Overall, the anti-IIa effect is 10 times greater than the anti-Xa effect (13), which is a testament to heparin's ability to form electrostatic bonds with other molecules.

Heparin's tendency for electrostatic binding extends to plasma proteins, endothelial cells, and macrophages. Heparin binding to plasma proteins reduces bioavailability, while heparin binding to endothelial cells and macrophages promotes clearance from the bloodstream. Variations in plasma protein levels add further to the variable anticoagulant effects of heparin.

PLATELET BINDING: Heparin binds to a protein (platelet factor 4) on platelets to form an antigenic complex that can trigger the formation of IgG antibodies. These antibodies can bind to platelets and trigger a platelet activation response, which leads to thrombosis and a consumptive thrombocytopenia. This *heparin-induced thrombocytopenia* is not a dose-dependent phenomenon, and can occur with the low doses of heparin used for thromboprophylaxis. (This condition is described in Chapter 13.)

Prophylactic Dosing

The activation of AT by heparin is a highly sensitive reaction, which means that low doses of heparin can inhibit thrombogenesis. *The dose of unfractionated heparin used for thromboprophylaxis is 5,000 units injected subcutaneously every 8–12 hours.* The 8-hour dosing schedule is popular for higher-risk situations, but most studies show no difference in efficacy (or risk of bleeding) with either regimen (14).

OBESITY: Standard doses of heparin are considered suboptimal in patients with obesity (body mass index or BMI ≥ 30 kg/m^2) because of the increased volume of drug distribution in obesity. As a result, higher doses of heparin have been recommended in obese patients, especially those with morbid obesity (BMI ≥ 40 kg/m^2). A popular regimen for patients with morbid obesity is 7,500 units every 8 hours, which has proven to be more effective in preventing VTE than the standard low-dose regimen (15).

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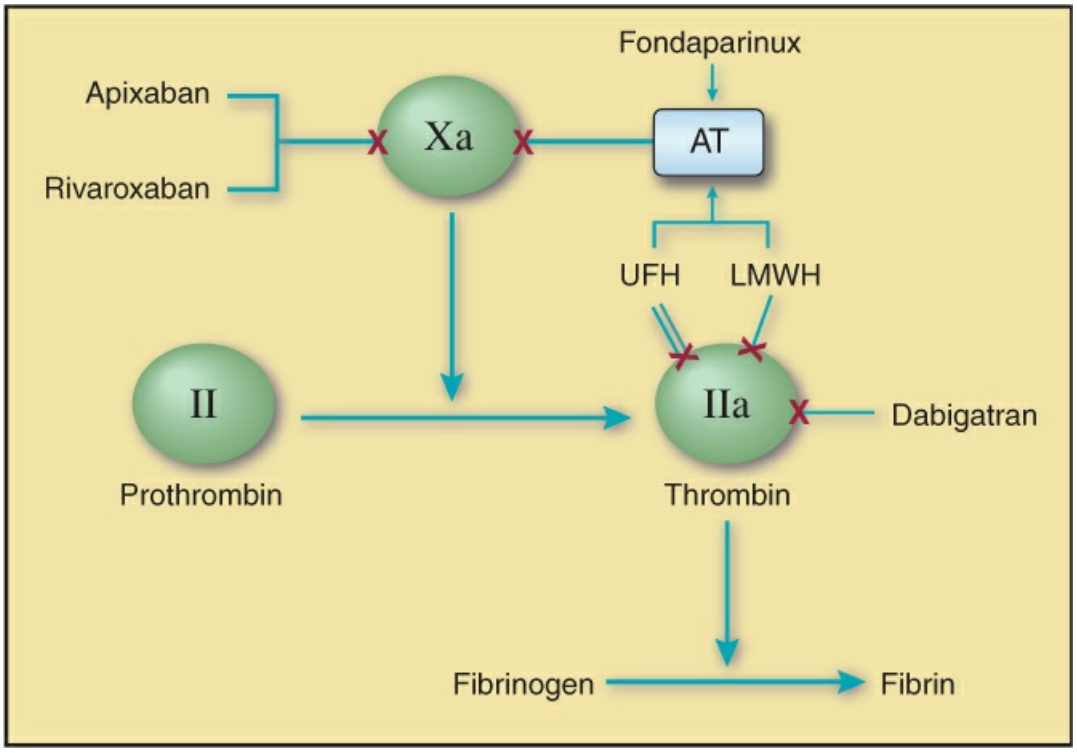


FIGURE 5.2 Mechanisms of anticoagulation with unfractionated heparin (UFH), low-molecular-weight heparin (LMWH) and the direct oral anticoagulants (DOACs). "X" denotes inhibition. See text for further explanation.

Low-Molecular-Weight Heparin (LMWH)

Low-molecular-weight heparin (LMWH) is produced by enzymatic cleavage of heparin molecules, which produces smaller molecules of more uniform size. (The average molecular weight of LMWH is 4,500 Daltons.) Like UFH, the LMWH molecules bind to AT and inhibit factor Xa (anti-Xa effect), but the shorter length of the molecules results in less binding to thrombin and reduces the anti-IIa effect (see Figure 5.2). This reduces the variability of anticoagulation seen with UFH, and gives LMWH a more predictable dose-response profile. This predictability is why laboratory measures of anticoagulation are not monitored during therapy with LMWH.

Other Advantages

LMWH does not bind as readily to plasma proteins, endothelial cells, macrophages, and platelets as UFH, and this gives LMWH the following advantages (13):

. As a result of reduced binding to plasma proteins, the bioavailability of LMWH is greater than that of UFH (90% vs 15–30%, respectively), which means that LMWH is a more potent anticoagulant than UFH (at equivalent doses).

. Reduced binding to endothelial cells and macrophages gives LMWH a longer duration of action than UFH, and this translates to less frequent dosing with LMWH.

. Reduced platelet binding by LMWH results in a lower risk of heparin-induced thrombocytopenia; i.e., the reported incidence of this complication is 0.2% with LMWH,

versus 2.6% with UFH (a ten-fold difference) (16).

In summary, *the advantages of LMWH over UFH include a more predictable and more potent anticoagulant response, less frequent dosing, and less risk of heparin-induced thrombocytopenia*.

Prophylactic Regimens

There are two LMWH preparations available for use in the United States: enoxaparin (Lovenox) and dalteparin (Fragmin), and Table 5.3 shows the prophylactic dosing regimens for each of these agents.

ENOXAPARIN: Enoxaparin was the first LMWH approved for use in the United States (in 1993), and numerous clinical studies have shown it to be both safe and effective for thromboprophylaxis. *The standard dose for thromboprophylaxis is 40 mg given by subcutaneous injection once daily, or 30 mg given twice daily for higher-risk patients*. The following dose adjustments deserve mention:

. Enoxaparin is cleared by the kidneys, and for patients with severe renal impairment (i.e., estimated creatinine clearance, or Cr CL, <30 mL/min), the standard prophylactic dose should be reduced to 30 mg once daily, or an alternative anticoagulant (e.g., UFH) should be used (17,18). The latter option seems to be the best choice when the Cr CL falls below 20 mL/min.

. For morbid obesity (BMI ≥ 40 kg/m^2) a dose of 40 mg every 12 hours has proven superior to standard prophylactic dosing (15). (See later for dosing adjustments related to bariatric surgery.)

Enoxaparin has proven to be both safe and effective for preventing VTE in all high-risk clinical settings (17), and it is more effective than UFH for orthopedic procedures involving the hip and knee (5), and in major trauma (6). It is currently the most popular agent for the prevention of VTE in hospitalized patients. (See later for recommendations in specific clinical settings.)

DALTEPARIN: The other LMWH, dalteparin, does not enjoy the popularity of enoxaparin, even though clinical studies have shown no difference in efficacy between dalteparin and enoxaparin for the prevention of VTE (18). *The prophylactic dose of dalteparin is 2,500–5,000 units given subcutaneously once daily.* Dalteparin has one notable advantage over enoxaparin; i.e., no dose reduction is required in patients with renal failure (19). The appropriate dose of dalteparin in obesity has not been adequately studied. One recommendation, based on an author's clinical experience, is a dose of 200 units/kg daily (actual body weight) (20).

Monitoring Anti-Factor Xa Levels

The adequacy of LMWH dosing can be evaluated by measuring the anti-factor Xa level in blood. This measure is not used routinely during LMWH therapy, but it can useful when the adequacy of fixed LMWH dosing is uncertain (e.g., in obesity). The recommended target range for prophylactic dosing is a peak anti-factor Xa level of 0.2–0.5 IU/mL (21).

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Fondaparinux

Fondaparinux (also known as "tiny heparin") is a synthetic analog of the pentasaccharide sequence on heparin that binds it to antithrombin III (AT). It is the first selective factor Xa inhibitor approved for clinical use, and has proven effective for thromboprophylaxis in medical patients (22) and in orthopedic procedures involving the hip and knee (23). *The dose of fondaparinux for thromboprophylaxis is 2.5 mg by subcutaneous injection once daily.* Fondaparinux has a half life of 17 hrs (which allows once-daily dosing), and is excreted by the kidneys. In patients with moderate-to-severe renal impairment (Cr CL of 20–50 mL/min), a reduced dose of 1.5 mg daily is recommended (24,25), and the drug should not be used when the Cr CL is <20 mL/min.

When to Use

Fondaparinux is recommended (along with UFH and LMWH) for hospitalized medical patients who require thromboprophylaxis (7), but it is not popular, and has not been studied in critically ill (ICU) patients. Fondaparinux can also be used for prophylaxis in patients with suspected heparin-induced thrombocytopenia (HIT), or a prior history of HIT (26). While it has been used in patients with confirmed HIT, both with and without thrombosis (26), the approved treatment for HIT is argatroban. (For more on HIT, see Chapter 13.)

PROPHYLAXIS WITH ORAL AGENTS

Thromboprophylaxis with oral agents is receiving increased interest as a result of the increase in extended prophylaxis after hospital discharge (see later). Warfarin was the original oral agent used for prophylaxis, but it is no longer favored because of the delayed onset of action (5–7 days), multiple drug interactions, and need for anticoagulation monitoring (with the INR). The newer oral anticoagulants, which include a direct thrombin inhibitor (dabigatran) and three direct factor Xa inhibitors (rivaroxaban, apixaban, and edoxaban), have a more rapid onset of action (1–4 hours) and do not require routine monitoring of anticoagulation status. (The actions of these oral agents are depicted in Figure 5.2.) These newer agents are collectively known as direct oral anticoagulants, or DOACs. At the present time, *prophylaxis with DOACs is recommended only for hip and knee replacement surgery* (2).

TABLE 5.3 Thromboprophylaxis with Oral Agents

Agent	Prophylactic Dosing Recommendations
Dabigatran (Pradaxa)	Standard: 110 mg initially, then 220 mg once daily. Obesity: Do not use if BMI ≥ 40 kg/m^2 or wt > 120 kg. Renal: Do not use if Cr CL < 30 mL/min.
Apixaban (Eliquis)	Standard: 2.5 mg twice daily. Obesity: Standard dosing for BMI ≥ 40 kg/m^2 or wt > 120 kg. Renal: Do not use if Cr CL < 15 mL/min.
Rivaroxaban (Xarelto)	Standard: 10 mg once daily. Obesity: Standard dosing for BMI ≥ 40 kg/m^2 or wt > 120 kg.

	Renal: Do not use if Cr CL < 15 mL/min.
Warfarin	Standard: 2.5–5 mg once daily; adjust dose to INR of 1.5–2.5. Obesity: No known dose adjustment. Renal: No dose adjustment.

From References 27–29. Cr CL = creatinine clearance, INR = international normalized ratio.

The dosing recommendations for prophylaxis with oral anticoagulants are summarized in Table 5.3, and some pertinent information about these agents is summarized below. (*Note:* Edoxaban is not included here because there is no approved dosage for thromboprophylaxis.)

- DOACs are contraindicated in patients with severe renal impairment (see Table 5.3); in this setting, warfarin is the only oral agent that can be used safely (27,28).
- Standard prophylactic doses of apixaban and rivaroxaban are equally effective in morbidly obese patients (i.e., BMI ≥ 40 kg.m^2 or wt. >120 kg) (29).
- The absorption of DOACs may be adversely affected by gastric bypass surgery, and (until this matter is studied further), these agents should probably not be used for prophylaxis after this procedure (29).
- All DOACs have proven superior to LMWH for thromboprophylaxis in hip and knee replacement surgery (2,27,28), but dabigatran and rivaroxaban have shown higher bleeding rates than LMWH in some studies (28), and apixaban may be the favored DOAC for these procedures (27).
- Rivaroxaban has been approved for thromboprophylaxis in patients with acute medical illness (30), but a meta-analysis of studies comparing DOACs and LMWHs in acutely ill medical patients shows equivalent efficacy but an increased risk of bleeding with DOACs (31).

The popularity of DOACs for venous thromboprophylaxis is very likely to skyrocket in the upcoming years, similar to what has happened with arterial thromboprophylaxis in atrial fibrillation (which is described in Chapter 19).

Aspirin

Although antiplatelet agents are typically not used for venous thromboprophylaxis, the most recent guidelines from the American Society of Hematology includes a conditional recommendation for aspirin as an effective alternative to anticoagulants for hip or knee replacement surgery (2). Although a dose is not specified, most studies involved low-dose aspirin (81 mg daily). This recommendation was based on studies that often included aspirin in conjunction with anticoagulant therapy; e.g., using anticoagulants in the early postoperative period, followed by aspirin for extended prophylaxis after hospital discharge (32). Therefore, the use of aspirin as the sole prophylactic agent after hip or knee arthroplasty seems unwarranted.

Neuraxial Procedures

Anticoagulant prophylaxis can promote hematoma formation from spinal punctures, or from the insertion and removal of intrathecal and epidural catheters, and these hematomas can compress the spinal cord and produce paralysis. To limit the risk of this complication, anticoagulant

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prophylaxis should be withheld before performing these procedures. The following are some guidelines for withholding anticoagulant prophylaxis before these procedures:

. For subcutaneous injections of unfractionated heparin, wait at least 12 hours after the last dose, and for once daily dosing of enoxaparin, wait at least 24 hours (9).

. For the DOACS, the recommended time for withholding anticoagulation is dependent on the renal function. With normal or moderately reduced renal function, the wait time is at least 48 hours for dabigatran and rivaroxaban, and at least 24 hours for apixaban (27). Longer wait times are recommended for more severe cases of renal impairment (see Reference 27 for wait times in relation to creatinine clearance).

MECHANICAL THROMBOPROPHYLAXIS

External compression of the lower extremities can be used to promote venous outflow from the legs and reduce the risk of VTE from venous stasis. This method of *mechanical thromboprophylaxis* is typically used as a replacement for anticoagulant regimens in patients who are bleeding or have a high risk of bleeding, but it can also be used as an adjunct to anticoagulant prophylaxis in a high-risk clinical setting (see later). There are two methods of external leg compression: graded compression stockings and intermittent pneumatic compression.

Graded Compression Stockings

Graded compression stockings (GCS), also known as thromboembolism-deterrent or TED stockings, are designed to create 18 mm Hg external pressure at the ankles and 8 mm Hg external pressure in the thigh (33). The resulting 10 mm Hg pressure gradient acts as a driving force for venous outflow from the legs. These stockings have been shown to reduce the incidence of VTE when used alone after major surgery (34). However, they are the least effective method of thromboprophylaxis, and are never used as sole means of preventing VTE in hospitalized patients.

Intermittent Pneumatic Compression

Intermittent pneumatic compression (IPC) has a dual antithrombotic effect because it promotes both venous flow and fibrinolysis (35). IPC devices typically have two or three inflatable bladders that are pressurized in sequence (from distal to proximal). The external leg pressure is about 35–40 mm Hg at the ankle, and 20 mm Hg at the thigh (33), and the compression cycle is typically 10 seconds (6 inflations per minute). (The applied pressure should never be greater than the diastolic blood pressure, to prevent arterial compression.) Venous outflow is enhanced not only by the pressure gradient along the leg, but also by the pumping action of the sequential bladder inflations.

The IPC method can be as effective as anticoagulant prophylaxis (36), and it also adds to the effectiveness of anticoagulant regimens (37). Therefore, it is no surprise that IPC is the standard method of mechanical prophylaxis in hospitalized patients. There are, however, a few shortcomings: i.e., the inflatable bladders restrict mobility and can macerate the skin, and the

repeated inflation and deflation of the bladders is often annoying for awake patients. Therefore, IPC should be discontinued as soon as it is no longer necessary.

Inferior Vena Cava Filters

Inferior vena cava (IVC) filters do not prevent deep vein thrombosis (DVT), but they reduce the risk of pulmonary embolism (by trapping thrombi that break loose from DVT in the legs). These devices are occasionally used in patients with DVT who cannot be anticoagulated, but otherwise they are not recommended for VTE prophylaxis. IVC filters are described in Chapter 22.

WHICH REGIMEN FOR WHICH PATIENT?

The following is a description of the recommended prophylactic regimens for specific clinical settings. Most of these recommendations are taken from the clinical practice guidelines listed at the end of the chapter (2–9). (When guidelines differ in their recommendations, the most recent guideline is usually chosen.) Please note the following specifics for these recommendations: (a) High-risk patients are those with a prior history of VTE, genetic risk factors for VTE, cancer, or multiple risk factors (e.g., advanced age, obesity, heart failure with leg edema); (b) The low-molecular-weight heparin (LMWH) is enoxaparin.

Special Circumstances

The following recommendations apply to all groups of patients or clinical settings:

. For patients with renal failure (creatinine clearance <15 mL/min), the only anticoagulants that can be used are unfractionated heparin (UFH) and coumadin.

. For patients with morbid obesity (BMI > 40 kg/m^2), consider adjusting the dose of the heparins (UFH and LMWH), as shown in Table 5.2.

. For patients with suspected heparin-induced thrombocytopenia (HIT), or a prior history of HIT, fondaparinux can be used for prophylaxis (and has been used in confirmed cases of HIT).

Major General Surgery

. The most recent guidelines recommend UFH or LMWH, with intermittent pneumatic compression (IPC) added in high-risk patients (2). Higher doses of UFH (5,000 units every 8 hours) and LMWH (enoxaparin, 30 mg twice daily) are popular for high-risk patients.

. Prophylaxis can be started preoperatively, but the last doses of UFH and LMWH should be at least 6 hrs and 12 hrs (respectively) before the surgical procedure. Postoperative prophylaxis can begin 6 hours after surgery for UFH, and 12 hours after surgery for LMWH.

. Prophylaxis is generally continued until full ambulation or hospital discharge, but for cancer surgery involving the abdomen and pelvis in high-risk patients, extended prophylaxis (with LMWH) for up to 4 weeks is favored (10).

Bariatric Surgery

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. The American Society for Metabolic and Bariatric Surgery favors LMWH plus IPC for prophylaxis, with an enoxaparin dose of 40 mg twice daily for a BMI of 30–49 kg/m^2, and 60 mg twice daily for a BMI ≥ 50 kg/m^2 (3).

. Following bariatric surgery, about 80% of postoperative VTEs occur after hospital discharge (3), so extended prophylaxis (with LMWH) is recommended for a total of about 2 weeks (3). (The duration of prophylaxis has not been adequately studied.)

Orthopedic Surgery

. For hip or knee replacement surgery, a DOAC (dabigatran, rivaroxaban, or apixaban) is favored over LMWH (2), with IPC added in the early postoperative period.

. For hip *fracture* surgery, LMWH is recommended (2,5). (DOACs have not been adequately studied in hip fracture surgery.)

. For major surgery involving the hip and knee, extended prophylaxis is recommended for up to 35 days after the procedure (2,5).

Neurosurgery

. For major neurosurgical procedures, mechanical prophylaxis with IPC is recommended.

. When the risk of postoperative bleeding has subsided, LMWH is added for high-risk patients, and for cancer-related surgery (2).

Cardiac Surgery

. For uncomplicated cases of cardiac surgery, mechanical prophylaxis with IPC is recommended (2,5).

. For complicated cardiac surgery, or high-risk patients, LMWH is added to IPC (5).

Major Trauma

. For solid organ injury or trauma to the extremities, prophylaxis with LMWH (enoxaparin at a dose of 40 mg SC every 12 hrs) should be started when bleeding is not a concern, optimally 12–24 hours after the injury, and mechanical prophylaxis with IPC (if tolerated) is added in high-risk patients (6).

. For traumatic brain injury, mechanical prophylaxis with IPC is used until the CT scan shows no bleeding, or progression of bleeding for 24 hours, then LMWH is added (enoxaparin at a dose of 30 mg SC every 12 hrs).

. For spinal trauma, prophylaxis with LMWH (enoxaparin at a dose of 30 mg SC every 12 hrs) should be started as soon as anticoagulation is considered safe, while mechanical prophylaxis with IPC should be started as soon as possible, and continued along with the LMWH (6).

. Monitoring anti-Xa levels is recommended in major trauma because of frequent variations in renal function.

Acute Medical Illness

. For high-risk patients with an acute medical illness (including all ICU patients), prophylaxis with LMWH is recommended, without adding mechanical prophylaxis (7). Extended prophylaxis after hospital discharge is not recommended (7).

A FINAL WORD

Riddle Me This

> If immobility is a risk factor for deep vein thrombosis (DVT),
> then why isn't DVT rampant in elderly nursing home residents
> who are bed-bound with leg contractures, and why isn't
> perpetual DVT prophylaxis used in people with paraplegia?

The perception that immobility or prolonged bed rest is a risk factor for DVT is an offshoot of Virchow's Triad, which identifies venous stasis as one of the principal risk factors for thrombosis. However, immobility is not necessarily venous stasis. The cardiovascular system is a closed system, and according to the law of conservation of energy and matter, the volumetric flow will be the same in all regions of a closed system. So if the cardiac output is normal, the arterial and venous flows will be the same (cardiac output equals venous return) and there will be no venous stasis with immobility or prolonged bed rest.

What triggers DVT in many of the high-risk hospitalized patients is probably inflammation (other than vascular injury in trauma and orthopedic patients), since the inflammatory and coagulation systems are closely linked. This would explain why in cases of spinal cord injury that result in paraplegia, the risk of DVT is high only for 3–4 weeks following the injury. Thereafter, thromboprophylaxis is discontinued, which implies that prolonged immobility alone is not a risk factor for DVT.

Why is this important? Because the perception that immobility predisposes to DVT is very likely a source of excessive use of anticoagulants for thromboprophylaxis.

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A L G r a w a n y

Chapter 6

Analgesia and Sedation in the ICU

To cure sometimes, to relieve often, and to comfort always.

Edward Livingston Trudeau, MD (1848−1915)

Contrary to popular perception, our principal function is not to save lives (since this is impossible on a consistent basis), but rather to *relieve pain and suffering*, and the ICU is the flagship of pain and suffering in hospitalized patients. In surveys of patients who have been discharged from the ICU, anxiety and unrelieved pain are the dominant recollections of the ICU stay (1). While it is not possible to completely eliminate the unpleasantness of the ICU stay, attention to patient comfort is always warranted, and surveys like one in Figure 6.1 (2) indicate that there is much work to be done.

This chapter describes the pharmacological approach to pain and stress relief in the ICU, with a focus on intravenous drug regimens and recommendations from expert commentaries on the subject (3−7). Nonpharmacological measures are not included here, but are available in a clinical practice guideline listed at the end of the chapter (4).

THE ICU EXPERIENCE

It is axiomatic to state that the ICU stay is a stressful experience for patients. Several stressors have been identified, including unrelieved pain and anxiety, dyspnea, the inability to communicate (in intubated patients), and disrupted sleep. The stress of a prolonged ICU stay can have far-reaching neuropsychiatric consequences, as post-traumatic stress disorder has been reported in patients who required prolonged periods of mechanical ventilation (8).

Pain in The ICU

Unrelieved pain is the most frequently cited stressor in the ICU (9), and the frequency of painful experiences is equivalent in surgical and medical ICUs (10).

Altered Pain Perception

Pain is generally defined as an unpleasant sensory experience (11). The degree of this unpleasantness (i.e., pain intensity) varies with the type and strength of the *nociceptive* (pain

producing) stimulus, and is also influenced by the presence and severity of disease. This latter condition applies to critically ill patients, as summarized in the following statements:

. The sensation of pain is magnified in critically ill patients, and this condition of *hypernociception* can transform a minor event into a painful experience.

. A common source of pain in ICU patients is the simple act of being turned in bed (9), and about 50% of patients experience pain at rest (usually in the lower back and legs), in the absence of a noxious stimulus (4,9,10).

. The heightened pain sensation in critically ill patients is attributed to systemic inflammation (via the production of prostenoids) and prolonged immobility.

Failure to recognize the exaggerated pain sensation in ICU patients is a source of inadequate pain control. However, this can be corrected with the use of pain assessment tools, like the ones described next.

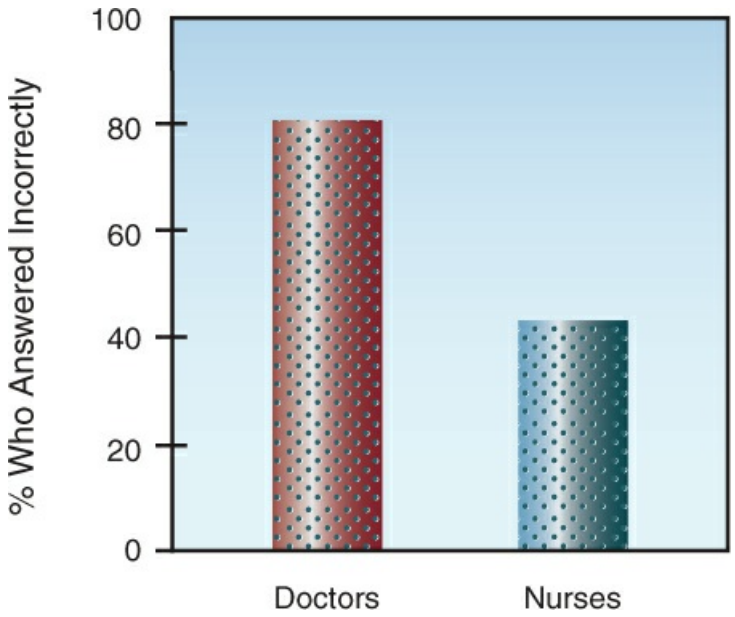


FIGURE 6.1 Percentage of resident physicians and ICU nurses who answered incorrectly when asked if diazepam (Valium) relieves pain.

Monitoring Pain

Routine assessments of pain intensity can be used to guide analgesic therapy. Because pain is a subjective sensation, the patient's self-assessment is the most accurate measure of pain intensity.

. When patients are capable of self-assessment, the *Numerical Ranking Scale* is recommended to quantify pain intensity (4). This is a horizontal scale with 10 equally spaced divider markings, numbered 1 (no pain) to 10 (maximum pain). The patient points to the numbered marking that corresponds to the intensity of pain. A score greater than 3 indicates unacceptable pain.

. When patients are not capable of self-assessment due to heavy sedation (e.g., during assisted ventilation), then physical signs can be used to assess pain. This is the basis for the *Behavioral Pain Scale* shown in Table 6.1 (12), which includes 12 individual behaviors separated into

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three categories: facial expression, arm flexion, and tolerance of mechanical ventilation. The final score ranges from 3 (no pain) to 12 (maximum pain). A score of 6 or higher indicates unacceptable pain.

Vital Signs

There is a tendency to look for changes in heart rate or blood pressure as evidence of pain, but this practice should be abandoned, because there is a poor correlation between pain intensity (as reported by patients) and changes in any of the vital signs (3,4).

TABLE 6.1 The Behavioral Pain Scale

Item	Description	Score
Facial Expression	• Relaxed • Partially tightened • Fully tightened • Grimacing	1 2 3 4
Upper Limbs	• No movement • Partially bent • Fully bent, fingers flexed • Permanently retracted	1 2 3 4
Compliance with Ventilation	• Tolerating ventilator • Coughing, but tolerating ventilator • Fighting ventilator • Unable to control ventilation	1 2 3 4
	Total Score	

Score	Interpretation
3	No Pain
≥6	Unacceptable pain
12	Maximal pain

From Reference 11.

ANALGESIA WITH OPIOIDS

The drugs described in this section are the most effective analgesics available (and hence the most popular drugs used for pain relief in the ICU). At the outset, a few definitions deserve mention:

. An *opiate* is a natural chemical derivative of opium.

. An *opioid* is a naturally-occurring or synthetic derivative of opium that acts by stimulating specific receptors (called *opioid receptors*) in the central nervous system and elsewhere. Stimulation of opioid receptors produces a variety of effects, including analgesia, sedation, euphoria, pupillary constriction, respiratory depression, bradycardia, constipation, nausea and vomiting, urinary retention, and pruritis.

. A *narcotic* is an outdated term for an opioid, but it has also been adopted by law enforcement agencies to refer to an opioid that is obtained illegally and used for non-medical purposes.

The terms opiate and opioid are often used interchangeably, but opioid is used here because it is a more accurate term for the drugs included in this section. In addition to their analgesic effects, opioids have the added benefit of producing mild sedation, but without amnesic effects (13). All opioids are metabolized in the liver, and the metabolites are excreted in the urine.

Drugs and Dosing Regimens

The opioids used most frequently in the ICU are morphine, fentanyl, and hydromorphone, and the recommended intravenous dosing for each drug is shown in Table 6.2 (3). It is important to emphasize that opioid dose requirements can vary widely in individual patients, and therefore *the effective dose of an opioid is determined by each patient's response, and not by the recommended dosing of the drug*.

TABLE 6.2 Commonly Used Intravenous Opioids

	Morphine	Hydromorphone	Fentanyl
Onset	5–10 min	5–15 min	1–2 min
Bolus Dosing	2–4 mg q 1–2 hr	0.2–0.6 mg q 1–2 hr	0.35–0.5 µg/kg q 0.5–1 hr
Infusion Rate	2–30 mg/hr	0.5–3 mg/hr	0.7–10 µg/kg/hr
PCA Demand (bolus) Lockout Interval	0.5–3 mg 10–20 min	0.1–0.5 mg 5–15 min	15–75 µg 3–10 min
Lipid Solubility	x	0.2x	600x
Active Metabolites	Yes	Yes	No
Histamine Release	Yes	No	No
Dose Adjustment for Renal Failure	↓ 50%	None	? Avoid†

Dosing recommendations from Reference 3. †From Reference 5.

Morphine

Morphine is the principal alkaloid of opium, and has been the traditional opioid used for pain control in the ICU. However, morphine has the following disadvantages that have curbed its popularity:

. Morphine has active metabolites that accumulate in renal failure. One metabolite (morphine 3-glucuronide) can produce agitation with myoclonus and seizures (14), while another metabolite (morphine-6-glucuronide) has more potent analgesic effects than the parent drug (15). To avoid accumulation of these metabolites, *the maintenance dose of morphine should be reduced by 50% in patients with renal failure* (16).

. Morphine promotes the release of histamine, which produces systemic vasodilation and a decrease in blood pressure (17). Hypotension is typically seen in patients with a

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hyperadrenergic state and increased peripheral vascular tone (3). Morphine-induced histamine release does not promote bronchoconstriction, as morphine doses of 1.5 mg/kg have been given to asthma patients without adverse consequences (18).

Fentanyl

Fentanyl is a synthetic opioid that has replaced morphine as the most popular opioid analgesic in ICUs (9). The advantages of fentanyl over morphine include the following:

. Fentanyl is 600 times more lipid soluble than morphine, and has a more rapid onset of action.

. Fentanyl does not promote histamine release (17), and thus has less risk of hypotension than morphine. The hemodynamic tolerance to fentanyl is a major source of its appeal in critically ill patients.

. Fentanyl has no active metabolites. (However, the parent drug can accumulate in renal failure, and some recommend avoiding fentanyl in patients with end-stage renal disease) (5).

One disadvantage of fentanyl is its lipophilicity, which promotes accumulation in the brain with continued usage.

Hydromorphone

Hydromorphone (Dilaudid®) is a semi-synthetic derivative of morphine that may produce more effective analgesia than morphine (19), and does not require dose alteration in renal failure. It seems to be favored for the treatment of cancer-related pain, but has no proven advantages over fentanyl for analgesia.

Remifentanil

Remifentanil is an ultra-short acting opioid that is 250 times more potent than morphine (6). The recommended intravenous dose begins with a loading dose of 1.5 µg/kg, and follows with a continuous infusion at 0.5–15 µg/kg/hr (3). Analgesic effects appear in about one minute, and are lost within 10 minutes after stopping the infusion. The short duration of action is a reflection of drug metabolism; i.e., remifentanil is broken down by nonspecific esterases in plasma (6). Since drug metabolism does not take place in the liver or kidneys, dose adjustments are not necessary in renal or hepatic failure.

Remifentanil has not been a popular drug for pain control in ICUs, but it has been used in patients with traumatic brain injury (20), where its short duration of action allows for frequent evaluations of neurologic status. The abrupt cessation of opioid activity can precipitate acute opioid withdrawal (6), which can be prevented by combining remifentanil with a longer-acting opioid.

Meperidine

Meperidine (Demerol®, Pethidine®) is an opioid analgesic that is no longer favored for pain control in the ICU because of the potential for *neurotoxicity*. Meperidine is metabolized in the liver to normeperidine, a metabolite that is slowly excreted by the kidneys (elimination half-life is 15–40 hours). Accumulation of normeperidine can produce central nervous system excitation, with agitation, myoclonus, delirium, and generalized seizures (21). Since renal dysfunction is

prevalent in ICU patients, accumulation of normeperidine is an unacceptable risk.

Patient-Controlled Analgesia

For patients who are awake and capable of drug self-administration, *patient-controlled analgesia* (PCA) can be an effective method of pain control, and may be superior to intermittent opioid dosing. The PCA method uses an electronic infusion pump that can be activated by the patient. When pain is sensed, the patient presses a button connected to the pump to receive a small intravenous bolus of drug. After each bolus, the pump is disabled for a mandatory time period called the *lockout interval*, to prevent overdosing. The opioid dosing regimens for PCA are shown in Table 6.2. The minimum lockout interval is determined by the time required to achieve peak drug effect (22). When writing orders for PCA, you must specify the initial loading dose (if any), the lockout interval, and the repeat bolus dose. PCA can be used alone or in conjunction with a low-dose opioid infusion.

Adverse Effects of Opioids

Opioids can produce a multitude of adverse effects;, and the following are the ones of most concern in ICU patients. (See also Chapter 52 for a description of opioid overdoses.)

Respiratory Depression

Opioids produce a centrally-mediated, dose-dependent decrease in respiratory rate and tidal volume, but *respiratory depression and hypoxemia are uncommon when opioids are given in the usual doses* (23). However, opioid doses that impair arousal can impair ventilation and promote hypercapnia (24). Patients with sleep apnea syndrome or chronic hypercapnia are particularly prone to respiratory depression from opioids.

Cardiovascular Effects

Opioid analgesia is often accompanied by decreases in blood pressure and heart rate, which are the result of decreased sympathetic activity and increased parasympathetic activity. (Morphine-induced hypotension may also be the result of histamine release.) Opioid effects on blood pressure are usually mild and well-tolerated, at least in the supine position (25). Decreases in blood pressure can be pronounced in patients with hypovolemia or heart failure (where there is an increased baseline sympathetic tone), or when opioids are given in combination with benzodiazepines (26). Opioid-induced hypotension is rarely a threat to tissue perfusion, and the blood pressure responds to intravenous fluids or small bolus doses of vasopressors.

Intestinal Motility

Opioids are well known for their ability to depress bowel motility via activation of opioid receptors in the GI tract. This is a source of troublesome constipation in cancer patients; in critically ill patients, impaired GI motility can promote reflux of enteral tube feedings into the oropharynx, creating a risk for aspiration pneumonia.

METHYLNALTREXONE: Methylnaltrexone is a peripheral opioid receptor antagonist that can block the GI effects of opioids without influencing the analgesic effects. This drug has proven successful in treating opioid-induced constipation in palliative care settings, but a preliminary

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study in ICU patients is not encouraging (27). In this study, intravenous methylnaltrexone in a dose of 12 mg daily (in adults weighing >60 kg) did not significantly increase laxation in patients with opioid-induced constipation.

Nausea and Vomiting

Opioids can promote vomiting via stimulation of the chemoreceptor trigger zone in the lower brainstem (24). All opioids are equivalent in their ability to promote vomiting, but vomiting induced by one opioid occasionally resolves when another opioid is used.

ANALGESIA WITH NON-OPIOIDS

Recent concerns about the rise in opioid dependency have prompted recommendations to reduce opioid use in hospitalized patients, if possible. To this end, the following analgesic agents should be considered as first-line agents or adjuvants to opioid analgesia in the ICU. The intravenous dosing regimens for these agents are presented in Table 6.3.

TABLE 6.3 Intravenous Non-Opioid Analgesia

Drug	Dosing Regimens and Comments
Acetaminophen	Dosing: 1 gram IV every 6 hr. Daily dose should not exceed 4 grams. Comment: Has no anti-inflammatory activity.
Ketorolac	Dosing: 30 mg IV every 6 hrs. Reduce dose by 50% for patients with renal failure, age ≥65 yrs, or body weight <50 kg. Comment: Serious complications (e.g., renal impairment) are uncommon with short-term use.
Ketamine	Dosing: 60–120 µg/hr, in combination with an opioid. Comment: Psychomimetic effects are not prominent at the doses used for opioid-sparing analgesia.

Acetaminophen

Acetaminophen was approved for intravenous use in 2010, and is intended for the short-term treatment of pain and fever in postoperative patients who are unable to receive the drug via the oral or rectal routes. The recommended dose is 1 gram IV every 6 hrs, with a maximum allowable dose of 4 grams daily (to prevent acetaminophen hepatotoxicity) (28). This dosing regimen has a documented opiate-sparing effect in postoperative patients (29). The major disadvantage of acetaminophen (in addition to the risk of hepatotoxicity) is the lack of an anti-inflammatory effect.

Ketorolac

Ketorolac is a nonsteroidal anti-inflammatory drug (NSAID) that produces analgesia without respiratory depression. The drug can be used alone for mild pain, but is more often used in combination with an opioid analgesic for moderate-to-severe pain. Ketorolac co-administration can reduce the opioid dose by 25−50% (30).

Dosing Regimen

Ketorolac can be given by intravenous (IV) or intramuscular (IM) injection, but IM injections of ketorolac can produce hematomas (31), so the IV route is preferred. The recommended dosing regimen for moderate-to-severe pain is 30 mg IV every 6 hrs (32). A 50% dose reduction (15 mg every 6 hrs) is recommended for patients with renal impairment, advanced age, and for those with a body weight <50 kg (32).

Risk

The beneficial actions of ketorolac and other NSAIDs are attributed to inhibition of prostaglandin production, but this also creates a risk for gastric mucosal injury and impaired renal function. These adverse effects are typically associated with excessive dosing or prolonged exposure to NSAIDs, and they are uncommon with short-term ketorolac use in the recommended doses (32–34).

Ketamine

Ketamine is a phencyclidine (PCP!) derivative that has anesthetic and analgesic properties, but it also has psychomimetic effects (e.g., dissociative reactions, hallucinations, and delirium after discontinuation), and these latter effects have limited its use as an analgesic outside the operating room. However, low doses of ketamine (60–120 μg/kg/hr) have been used in combination with opioids (opioid-sparing effect) without troublesome psychomimetic side effects (6).

Other notable effects of ketamine include increases in blood pressure and heart rate (sympathomimetic effects), bronchodilation, nausea and vomiting, and hypersalivation (6). The sympathomimetic effects may be advantageous in patients with circulatory shock, while the bronchodilation is desirable in patients with asthma. However, the non-analgesic effects of ketamine are diminished in the dose range used for opioid-sparing analgesia. Ketamine is generally not advised for patients with myocardial ischemia (6).

Neuropathic Pain

Non-opioid analgesia is usually required for neuropathic pain (e.g., from diabetic neuropathy), and the recommended drugs for this type of pain are *gabapentin* and *carbamazepine* (3). Both drugs must be given enterally. Effective drug doses vary in individual patients, but typical doses are 600 mg every 8 hrs for gabapentin, and 100 mg every 6 hours for carbamazepine.

ANXIETY IN THE ICU

Anxiety and related disorders (agitation and delirium) are observed in as many as 85% of patients in the ICU (35). These disorders can be defined as follows:

. *Anxiety* is characterized by exaggerated feelings of fear or apprehension that are sustained by internal mechanisms more than external events.

. *Agitation* is a state of anxiety that is accompanied by increased motor activity.

. *Delirium* is an acute confusional state that may, or may not, have agitation as a component. Although delirium is often equated with agitation, there is a hypoactive form of delirium that is

characterized by lethargy. (Delirium is described in more detail in Chapter 45.)

The common denominator in these disorders is the *absence of a sense of well-being*.

Sedation

Sedation is the process of relieving anxiety and establishing a state of calm. This process includes general supportive measures (like frequent communication with patients and families), and drug therapy. Several drugs are available for sedation in ICU patients, including benzodiazepines (e.g., midazolam), propofol, dexmedetomidine, and haloperidol.

Monitoring Sedation

The routine use of sedation scales is instrumental in achieving effective sedation in the ICU. The sedation scales that are most reliable for ICU patients are the Sedation-Agitation Scale (SAS) and the Richmond Agitation-Sedation Scale (RASS) (3), which is shown in Table 6.4 (36). The RASS includes 4 possible scores for progressive agitation (+1 to +4) and 5 possible scores for progressive sedation (–1 to –5). The optimal RASS score is zero (alert and calm). The added advantage of RASS is the ability to monitor serial changes in a patient's mental state (37). This latter feature allows the RASS score to be used as the end-point of sedative drug therapy. (Sedative drug infusions can be titrated to achieve a RASS score of –1 to –2, which represents light sedation.)

TABLE 6.4 The Richmond Agitation-Sedation Scale

Score	Term	Description
+4	Combative	Overly combative or violent; immediate danger to staff
+3	Very agitated	Pulls on or removes tube(s), catheter(s), or aggressive behavior
+2	Agitated	Frequent non-purposeful movement or patient-ventilator asynchrony
+1	Restless	Anxious or apprehensive but movements not aggressive or vigorous
0	Alert & calm	
−1	Drowsy	Not fully alert, but awakens for >10 sec, with eye contact, to voice
−2	Light sedation	Briefly awakens (<10 sec), with eye contact to voice
−3	Moderate sedation	Any movement (but no eye contact) to voice
−4	Deep sedation	No response to voice, but movement to physical stimulation
−5	Unarousable	No response to voice or physical stimulation

To determine the RASS, proceed as follows:

Step 1 Observation: Observe the patient without interaction. If patient is alert, assign the appropriate score (0 to +4). If patient is not alert, go to Step 2.

Step 2 Verbal Stimulation: Address the patient by name in a loud voice and ask the patient to look at you. Can repeat once if necessary. If patient responds to voice, assign the appropriate score (−1 to −3). If there is no

response, go to Step 3.

Step 3 Physical Stimulation: Shake the patient's shoulder. If there is no response, rub the sternum vigorously. Assign the appropriate score (−4 to −5).

From Reference 36.

BENZODIAZEPINES

Benzodiazepines have been the traditional sedative agents used in the ICU, but they are gradually losing ground to other sedatives because of drug accumulation with prolonged use, and an increased risk of ICU-related delirium (see later). There are two benzodiazepines that are favored for intravenous delivery: midazolam (Versed®) and lorazepam (Ativan®). Diazepam (Valium®) is generally avoided because the long half-life (90 hours) and presence of active metabolites promotes prolonged sedation, especially with renal impairment. The intravenous dosing regimens for midazolam and lorazepam are shown in Table 6.5 (3).

Midazolam

Midazolam is a short-acting drug that is favored for procedural sedation. Sedative effects are apparent within 5 minutes, and they last for 1–2 hours. Because of the short-lived effect, midazolam can be given as a continuous IV infusion for more prolonged sedation. However, midazolam is highly lipid soluble, and will accumulate in the central nervous system when infused continuously. To avoid prolonged sedation from drug accumulation, *midazolam infusions should be limited to ≤48 hrs* (4).

Midazolam is metabolized by the cytochrome P450 enzyme system in the liver, and drugs that interfere with this enzyme system (e.g., diltiazem, erythromycin) can potentiate the sedative effects. Midazolam has one active metabolite that is cleared by the kidneys, and renal impairment can add to the tendency for prolonged sedation (3,4).

Lorazepam

Lorazepam has a rapid onset of action, like midazolam, but the effect lasts longer (up to 6 hours). Continuous infusions of lorazepam can also lead to drug accumulation and prolonged sedation, but this effect is less pronounced (38), because lorazepam is less lipid soluble than midazolam, and has no active metabolites. However, there is a more serious risk associated with lorazepam infusions: i.e., propylene glycol toxicity (see next).

Propylene Glycol Toxicity

Propylene glycol is added to certain intravenous drug preparations to enhance drug solubility in plasma, and intravenous lorazepam has the highest concentration of propylene glycol of all drug preparations. Propylene glycol is converted to lactic acid in the liver, and excessive intake can produce a clinical syndrome characterized by a metabolic (lactic) acidosis, altered mentation, systemic inflammation, and renal failure (39). The risk of this "toxidrome" emerges when the lorazepam infusion rate exceeds the upper limit of the recommended rate (0.1 mg/kg/hr), or is >10 mg/hr, for longer than 48 hours (39,40).

TABLE 6.5 Sedation with Intravenous Benzodiazepines		
Feature	**Midazolam**	**Lorazepam**
Bolus Dose	0.01–0.05 mg/kg	0.02–0.04 mg/kg
Onset of Action	3–5 min	5–15 min
Duration	1–2 hr	2–6 hr
Continuous Infusion	0.02–0.1 mg/kg/hr	0.01–0.1 mg/kg/hr
Lipid Solubility	+++	++
Concerns	Active Metabolites	Propylene Glycol Toxicity

Dosing recommendations from Reference 3.

OSMOLAL GAP: Plasma levels of propylene glycol are not routinely available, but an elevated osmolal gap (i.e., >10 mosm/kg H_2O) has shown a good correlation with propylene glycol accumulation (41). (Lactate levels are unreliable for predicting either propylene glycol levels or the risk of clinical symptoms.) Therefore, monitoring the osmolal gap is recommended when the lorazepam infusion rate is ≥0.1 mg/kg hr, or >10 mg/hr, for 48 hours (39–41). (See Chapter 31 for more on the osmolal gap, and how to calculate the plasma osmolality.)

Advantages

The advantages of sedation with benzodiazepines include the following:

. The rapid onset of action makes benzodiazepines well-suited for the acute management of anxiety and agitation.

. Benzodiazepines cause a dose-dependent anterograde amnesia that is distinct from the sedative effect (42). Short-term memory is less affected than long-term memory. This is considered advantageous because it eliminates memories of stressful experiences, and it may explain why as many as 40% of patients who leave the ICU have no recollection of events that occurred while in the ICU (1,43).

. Benzodiazepines are the sedatives of choice for the management of alcohol and opioid withdrawal.

Disadvantages

The major disadvantages of benzodiazepines are drug accumulation with prolonged use, and a perceived tendency to promote delirium.

Prolonged Sedation

The major disadvantage of benzodiazepines is their tendency to accumulate and produce excessive sedation and delayed awakening, which can delay weaning from mechanical ventilation and prolong the ICU stay. As mentioned earlier, this is more of a problem with midazolam than lorazepam; e.g., in one study of prolonged sedation in ventilator-dependent patients, the time to emerge from sedation was 30 hours for midazolam and 4–5 hours for

lorazepam (38). Lorazepam is thus the benzodiazepine-of-choice for prolonged periods of continuous-infusion sedation (e.g., in ventilator-dependent patients) (4).

Delirium

Clinical studies have shown a relationship between benzodiazepine use and delirium in ICU patients (4,44). The link may be gamma-amino-butyric-acid (GABA), the principal inhibitory neurotransmitter in the brain; i.e., benzodiazepines act by binding to GABA receptors in the brain, and GABA-mediated neurotransmission is also involved in the development of delirium (44). This is supported by evidence that sedation with drugs that do not involve GABA receptors is less likely to be accompanied by delirium (44).

Preventive Measures

The following strategies are available for reducing the risks of oversedation and/or delirium associated with the prolonged use of benzodiazepines:

. Daily interruption of benzodiazepine infusions, known as a "sedation holiday", is an effective measure to hasten weaning from mechanical ventilation (4,45).

. Maintaining light sedation by titrating benzodiazepine dosing to a specific target on one of the sedation scales has been shown to promote weaning from mechanical ventilation and reduce the risk of delirium (4,44).

. Avoiding benzodiazepines for prolonged sedation during mechanical ventilation, if possible, is the recommendation of the most recent guidelines on sedation in the ICU (4).

SEDATION WITH RAPID-AROUSAL AGENTS

Concerns about the risks associated with benzodiazepines has prompted a shift in attention to two agents that allow rapid arousal from sedation: propofol and dexmedetomidine.

Propofol

Propofol (Deprivan®) is a rapidly-acting, GABAergic sedative that was introduced for the induction of general anesthesia, and has subsequently become a popular agent for sedation during mechanical ventilation. A profile of this drug is presented in Table 6.6

TABLE 6.6 Sedation with Rapid Arousal Agents

Feature	Propofol	Dexmedetomidine
Loading Dose	5 µg/kg/min over 5 min	1 µg/kg over 10 min
Onset of Action	1−2 min	5−10 min
Maintenance Infusion	5−50 µg/kg/min	0.2–1.5 µg/kg/hr
Time to Arousal	10−15 min†	6−10 min
Respiratory Depression	Yes	No

Adverse Effects	• Hypotension • Hyperlipidemia • Propofol Infusion Syndrome	• Hypotension • Bradycardia

†Arousal time is influenced by depth of sedation and duration of the infusion (see Table 6.7). Dosing recommendations from Reference 3.

Actions

Propofol has sedative, amnesic, and anticonvulsant effects, but no analgesic effects (46). A single intravenous bolus produces sedation within 1–2 minutes, and the drug effect lasts 5–8 minutes (46). Prolonged sedation requires a continuous drug infusion, and termination of the infusion can lead to arousal in only 10–15 minutes (46). However, the arousal time is dependent on the depth of sedation and the duration of the continuous infusion. This is demonstrated in Table 6.7, which shows the arousal times in ICU patients who were maintained at different levels of sedation (light or deep sedation) during propofol infusions that varied from one day to 14 days (47). Note that arousal times were longer with deep sedation and longer infusion times, and that the depth of sedation had the most profound effect on the time to arousal. These result highlight the importance of maintaining light sedation with propofol (if possible) to ensure more rapid emergence from sedation.

Preparation and Dosage

Propofol is suspended in a 10% lipid emulsion to enhance solubility in plasma. This lipid emulsion is almost identical to the 10% lipid formulation used in parenteral nutrition formulas, and it has a caloric density of 1 kcal/mL (which should be included as part of the daily caloric intake). *Propofol dosing is based on ideal rather than actual body weight*, and no dose adjustment is required for renal failure or hepatic insufficiency (46).

Adverse Effects

The most common side effects of propofol are respiratory depression and hypotension (5,48). The risk of respiratory depression mandates that only patients receiving mechanical ventilation can be sedated with propofol. The hypotension is attributed to systemic vasodilation (5), and can be profound in conditions like hypovolemia and heart failure, where blood pressure is maintained by systemic vasoconstriction. Loading doses of propofol are not advised in patients who are hemodynamically unstable (3).

Other notable consequences of propofol sedation include hypertriglyceridemia (attributed to the lipid emulsion used to deliver the drug) (48), necrotizing pancreatitis (rare) (49), anaphylactoid reactions (uncommon), a greenish hue in urine (from harmless phenolic metabolites) (46), and the life-threatening condition described next.

PROPOFOL INFUSION SYNDROME: Propofol infusion syndrome is an uncommon condition characterized by the abrupt onset of systolic heart failure, unrelenting lactic acidosis, rhabdomyolysis, acute renal failure, and eventual multiorgan failure (50). The etiology is unclear, but may be the result of propofol-induced inhibition of the mitochondrial electron transport chain (which produces high-energy ATP molecules). This syndrome typically occurs during high-dose propofol infusions (>80 µg/kg/min, or >5 mg/kg/hr) for longer than 48 hours (51), but it has also been reported at moderate infusion rates (36 µg/kg/min) (52). The prevalence

of this condition is unclear, since many of the manifestations are shared by other severe illnesses, but an incidence of 3% has been reported (52). Management involves general supportive care, and ECMO has been used in a few cases (50). The mortality rate is 30–35% (52,53).

TABLE 6.7 Factors that Influence Emergence from Propofol Sedation

Infusion Time	Arousal Time	
	Light Sedation	Deep Sedation
24 hours	13 minutes	25 hours
72 hours	34 minutes	59 hours
7 days	3.3 hours	71 hours
14 days	3.4 hours	74 hours

From Reference 47.

Dexmedetomidine

Dexmedetomidine (Precedex®) is an alpha-2 adrenergic agonist that has sedative, amnesic, and mild analgesic effects, and does not depress ventilation (54). A brief profile of the drug is presented in Table 6.6. The most distinguishing feature of dexmedetomidine is the type of the sedation it produces, which is described next.

Cooperative Sedation

The sedation produced by dexmedetomidine is unique because *arousal is maintained, despite deep levels of sedation*. Patients can be aroused from sedation without reducing or halting the drug infusion, and when arousal is no longer necessary, the patient will spontaneously return to the prior state of sedation. This has been called *cooperative sedation* (6), and it is more like a sleep state than the drug-induced stupor produced by the benzodiazepines and propofol. In fact the EEG changes in dexmedetomidine-induced sedation are similar to the changes that occur in natural sleep (6).

Advantages

Sedation with dexmedetomidine is well-suited for situations where temporary arousal is desirable, such as weaning attempts during mechanical ventilation, or periodic evaluations of mental status in stroke or traumatic brain injury. Other advantages include the following:

. Sedation with dexmedetomidine obviates the need for periodic or daily interruption of sedative infusions (the sedation holiday), thereby allowing patients to be maintained at constant level of sedation.

. Since dexmedetomidine does not depress ventilation, it can be used in patients who are not ventilator-dependent.

. The analgesic effects of dexmedetomidine can be advantageous for reducing opioid requirements (opioid sparing effect).

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. Delirium is less frequent with dexmedetomidine compared to benzodiazepines or propofol (55). In addition, dexmedetomidine is the drug-of-choice for the treatment of ICU-related delirium (4). This is a significant advantage for dexmedetomidine, since about 40% of ICU patients above the age of 65 will develop delirium (56).

Because of these advantages, dexmedetomidine is steadily growing in popularity for the sedation of critically ill patients, and it is considered superior to benzodiazepines for sedation during mechanical ventilation (4).

Disadvantages

. Dexmedetomidine produces dose-dependent decreases in heart rate, blood pressure, and circulating norepinephrine levels (sympatholytic effect) (6), and this effect is particularly marked in conditions where heart rate and blood pressure are supported by increased sympathetic activity; e.g., heart failure with reduced ejection fraction (57).

. Dexmedetomidine can have a withdrawal syndrome characterized by signs of sympathetic overactivity (i.e., tachycardia, hypertension, and agitation) (58). Dexmedetomidine withdrawal is reported in 30% of patients (58), and is unrelated to the infusion rate or duration.

ANTIPSYCHOTIC AGENTS

Several antipsychotic agents are used for sedation in hospitalized patients, including Ziprasidone (Geodon®), quetiapine (Seroquel®), risperidone (Risperdal®), and haloperidol (Haldol®). The following presentation focuses on haloperidol, which is the only antipsychotic agent that can be given intravenously, and has a long history of treating agitation and delirium in hospitalized patients.

Haloperidol

Haloperidol is a dopamine receptor antagonist that promotes sedation without troublesome respiratory depression or cardiovascular compromise (although hypotension is a risk in hypovolemic patients) (59). The principal use of haloperidol has been the treatment of agitation and delirium that are not the result of alcohol or drug withdrawal. As mentioned earlier, dexmedetomidine is considered the drug of choice for hospital-related delirium (4), but haloperidol still has a role for patients who do not tolerate dexmedetomidine because of bradycardia or hypotension.

TABLE 6.8 Intravenous Halperidol for Sedation†

1. Begin with 5 mg of haloperidol as an intravenous bolus.
2. For severely agitated or disruptive patients, add midazolam (1 mg) for a more rapid onset of sedation.
3. Wait 15 minutes.
4. If no response, double the dose of haloperidol (10 mg IV).
5. If no response after the second dose, switch to another agent.
6. If the response is satisfactory, maintain sedation with periodic doses of haloperidol (at 25% of the effective dose)

given at 6–hour intervals.

†Do not use if the corrected QT interval (QTc) is >500 msec.

Dosing Recommendations

Haloperidol is given in intravenous bolus doses for the acute management of agitation and delirium. Following a single intravenous dose, the sedative effect begins after 10–20 minutes, and it lasts for 3−4 hours (59).

A dosing regimen for intravenous haloperidol is outlined in Table 6.8. (*Caveat:* Do not use haloperidol if the corrected QT interval (QTc) on the ECG is >500 msec.) The initial dose of haloperidol is typically 5 mg (60). Since haloperidol is not a rapidly-acting drug, midazolam (1−2 mg) can be given with the haloperidol to achieve more rapid control in patients who are severely agitated or disruptive. If the initial dose of haloperidol does not produce the desired level of sedation after 20 minutes, a second dose is given at twice the strength of the initial dose (10 mg). If the second dose of haloperidol is ineffective, then switch to another agent. If the response to haloperidol is satisfactory, then sedation is maintained with periodic doses (at 25% of the effective initial dose) at 4−6 hour intervals (61).

Adverse Effects

The following are the notable risks associated with haloperidol:

. *Extrapyramidal reactions* (e.g., rigidity, spasmodic movements) are dose-related side effects when haloperidol is given orally, but these reactions *are uncommon when haloperidol is given intravenously* (for unclear reasons) (62).

. The most noted risk with haloperidol is prolongation of the QT interval on the ECG, which can trigger a polymorphic ventricular tachycardia known as *torsade de pointes* (which is described in more detail in Chapter 19). This arrhythmia has been reported in 3–4% of patients who received intravenous haloperidol (63), and most cases occurred when the corrected QT interval (QTc) was >500 msec, or the daily dose of haloperidol exceeded 35 mg. Therefore, a QTc >500 msec is considered a contraindication to intravenous haloperidol.

. One of the more egregious (and often overlooked) complications of haloperidol is the *neuroleptic malignant syndrome*, which is an idiosyncratic reaction that presents with hyperpyrexia, severe muscle rigidity, and rhabdomyolysis. This condition has been reported in ICU patients receiving intravenous haloperidol (64), and it can be fatal if left unnoticed. (See Chapter 43 for a more detailed description of this condition.)

A FINAL WORD

From Opiophobia to Opiophilia, and Back Again

Opiophobia

The term “opiophobia” was introduced in the 1980s (65) to describe the reluctance of physicians to prescribe opioids for pain relief, based on the fear of promoting opioid abuse in the community. The general consensus at that time was that pain was undertreated, and opiophobia

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was tagged as the culprit. To correct this, The American Pain Society named pain as the "fifth vital sign" (to increase awareness of untreated pain) (66) and to encourage the use of opioids, emphasis was placed on a single study involving 11,000 patients who received opioids during their hospital stay, which showed that only four of the patients developed an opioid addiction (67). (This study was actually a letter to the editor.)

Opiophilia

The efforts to encourage opioid use that began in the 1990s were a smashing success, and by 2010, opioid prescription hit a peak of 81 prescription for every 100 Americans (68)! This "opiophilia" has, of course, been blamed for the current "epidemic" of opioid abuse and opioid-related deaths. In the twenty years from 1999 to 2019, there were an estimated 500,000 deaths from opioid overdose (69). Of interest, opioid prescribing has declined in recent years, but the opioid death toll continues to rise.

Back to Opiophobia

The current opioid abuse problem has prompted efforts to curb the clinical use of opioids. The aim is to foster a reluctance to prescribe opioids for pain relief, based on the fear of promoting opioid abuse in the community, which is opiophobia (again)! *And so it goes*.

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Section III

PHYSIOLOGICAL MONITORING

Not everything that can be counted counts, and not everything that counts can be counted.

Albert Einstein

Chapter 7

Oximetry and Capnography

The killing vice of a young doctor is intellectual laziness.

Sir William Osler (*a*)

The introduction of optical techniques for the noninvasive and continuous monitoring of oxyhemoglobin saturation in blood (oximetry) and carbon dioxide levels in exhaled gas (capnography) is simply the most useful innovation in critical care monitoring since the advent of intensive care units (in this author's opinion). The reach of "pulse" oximetry has extended beyond the ICU, and even beyond the hospital, with the pulse oximetry measurement being proposed as the fifth vital sign (1), while exhaled CO_2 monitoring has become an indispensable tool for ensuring the safety of procedural sedation, and for predicting the outcome of cardiopulmonary resuscitation.

Despite the ubiquity and promise of pulse oximetry and capnography, there is evidence that the understanding of these techniques is far from satisfactory (2). The information in this chapter should help to correct this deficiency.

OXIMETRY: METHODOLOGY

Molecules absorb light at wavelengths that are specific for each molecule. This is the basis for the technique of *spectrophotometry*, which uses light waves to identify the molecular composition of a medium. The application of this technique to the detection of hemoglobin in its different forms is known as *oximetry*.

Light Absorption by Hemoglobin

Hemoglobin (like all proteins) changes its structural configuration when it participates in a chemical reaction, and each of the hemoglobin configurations has a distinct pattern of light absorption. This is demonstrated in Figure 7.1 (3). Four different forms of hemoglobin are represented: oxygenated hemoglobin (HbO_2) deoxygenated hemoglobin (Hb), methemoglobin (metHb) and carboxyhemoglobin (COHb). In the red region of the light spectrum (represented by the wavelength of 660 nm), oxygenated hemoglobin (HbO_2) does not absorb light as well as deoxygenated hemoglobin (Hb), which is why oxygenated blood is more intensely red than

deoxygenated blood. The opposite is true in the infrared region of the spectrum (represented by the wavelength of 940 nm), where HbO_2 absorbs light more effectively than Hb. Based on these absorption patterns, two wavelengths of light (660 nm and 940 nm) can be used to identify oxygenated and deoxygenated hemoglobin.

Early Oximetry

Oximetry was introduced in the 1940s to detect hypoxemia in fighter pilots. The early use of oximetry involved the transmission of red and infrared light beams through the earlobes, but this methodology had two major shortcomings: (a) the transmission of light was influenced by factors other than hemoglobin (e.g., skin pigments, tissue thickness), and (b) it was not possible to differentiate between the hemoglobin in arteries and veins. As a result of these problems, oximetry failed to gain acceptance as a monitoring tool until the 1970s, when *pulse oximetry* was introduced.

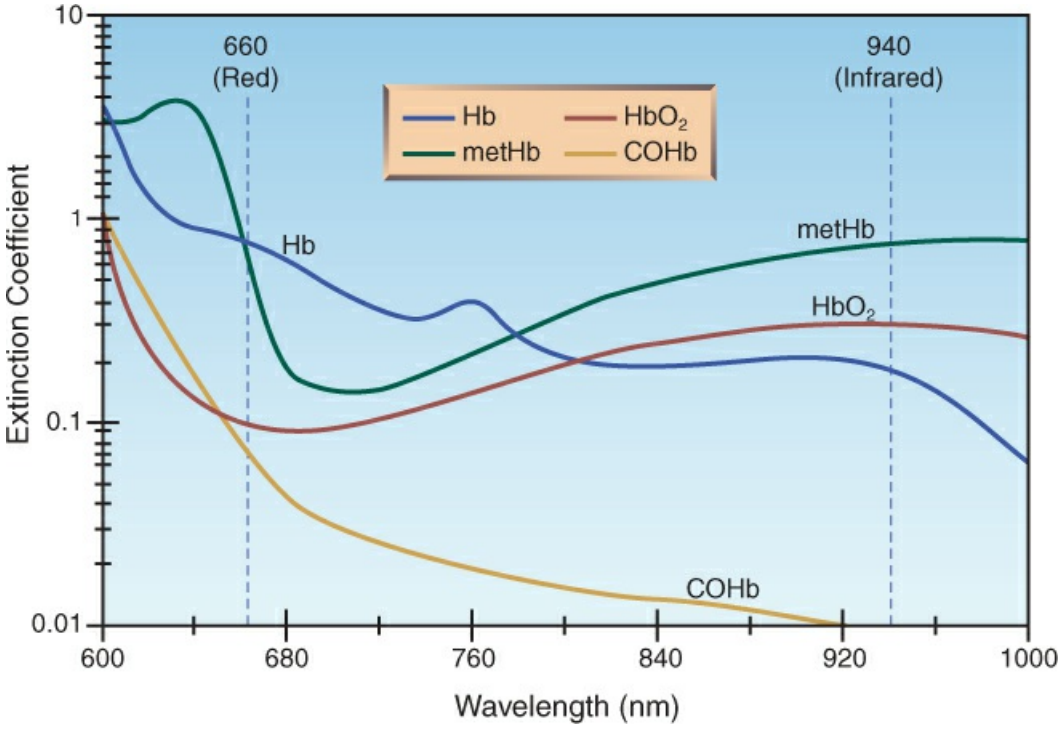


FIGURE 7.1 Light absorption by oxygenated hemoglobin (HbO_2), deoxygenated hemoglobin (Hb), carboxyhemoglobin (COHb), and methemoglobin (metHb) The vertical lines represent the two wavelengths of light (660 nm and 940 nm) used by pulse oximeters. Adapted from Reference 3.

Pulse Oximetry

When a light beam passes through a pulsating artery, the phasic changes in arterial blood volume create pulsatile variations in the intensity of the transmitted light beam. This is demonstrated in the "photoplethysmogram" in Figure 7.2, which shows that the increase in arterial blood volume during systole is associated with a decrease in the transmission of light through the artery. Therefore, restricting the analysis of red and infrared light transmission to the systolic fraction of the photoplethysmogram (A/B in Figure 7.2) will focus the analysis on arterial blood, thereby

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reducing or eliminating errors due to light absorption by non-pulsatile elements (e.g., hemoglobin in veins). This is the basic principle for pulse oximetry (3,4).

Fingertip Pulse Oximetry

Pulse oximetry is typically performed with a fingertip probe like the one in Figure 7.3. One side of the probe contains two light-emitting diodes that emit monochromatic light at wavelengths of 660 nm and 940 nm. These light waves pass through the finger and are sensed by a photodetector on the opposite side of the probe. These probes also generate a pulsatile photoplethysmography tracing, which is shown at the bottom of the figure. (Note that the tracing is reversed to resemble an arterial pressure waveform.) The intensity of light transmission during the systolic portion of the photoplethysmography tracing is used as a reflection of the deoxygenated hemoglobin (Hb, at 660 nm) and oxygenated hemoglobin (HbO_2, at 940 nm) in arterial blood. The ratio of HbO_2 to total hemoglobin (HbO_2 + Hb) is then used to define the fraction of hemoglobin that is saturated with oxygen. The resulting "pulse oximeter saturation" (SpO_2) is expressed as a percentage:

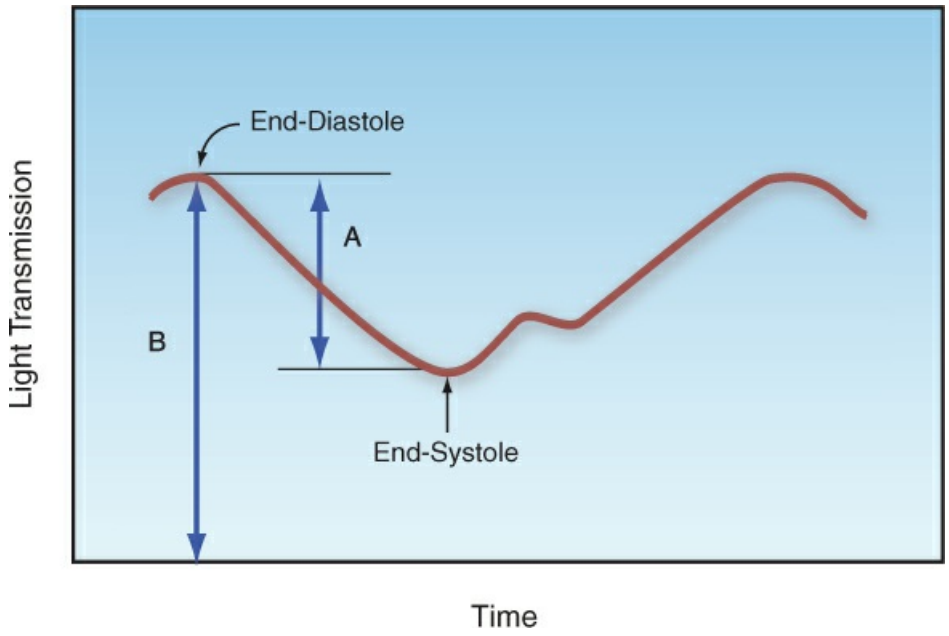


FIGURE 7.2 Photoplethysmography tracing showing the changes in light transmission resulting from the pulsatile change in arterial blood volume. A = systolic amplitude, B = baseline transmission. See text for further explanation.

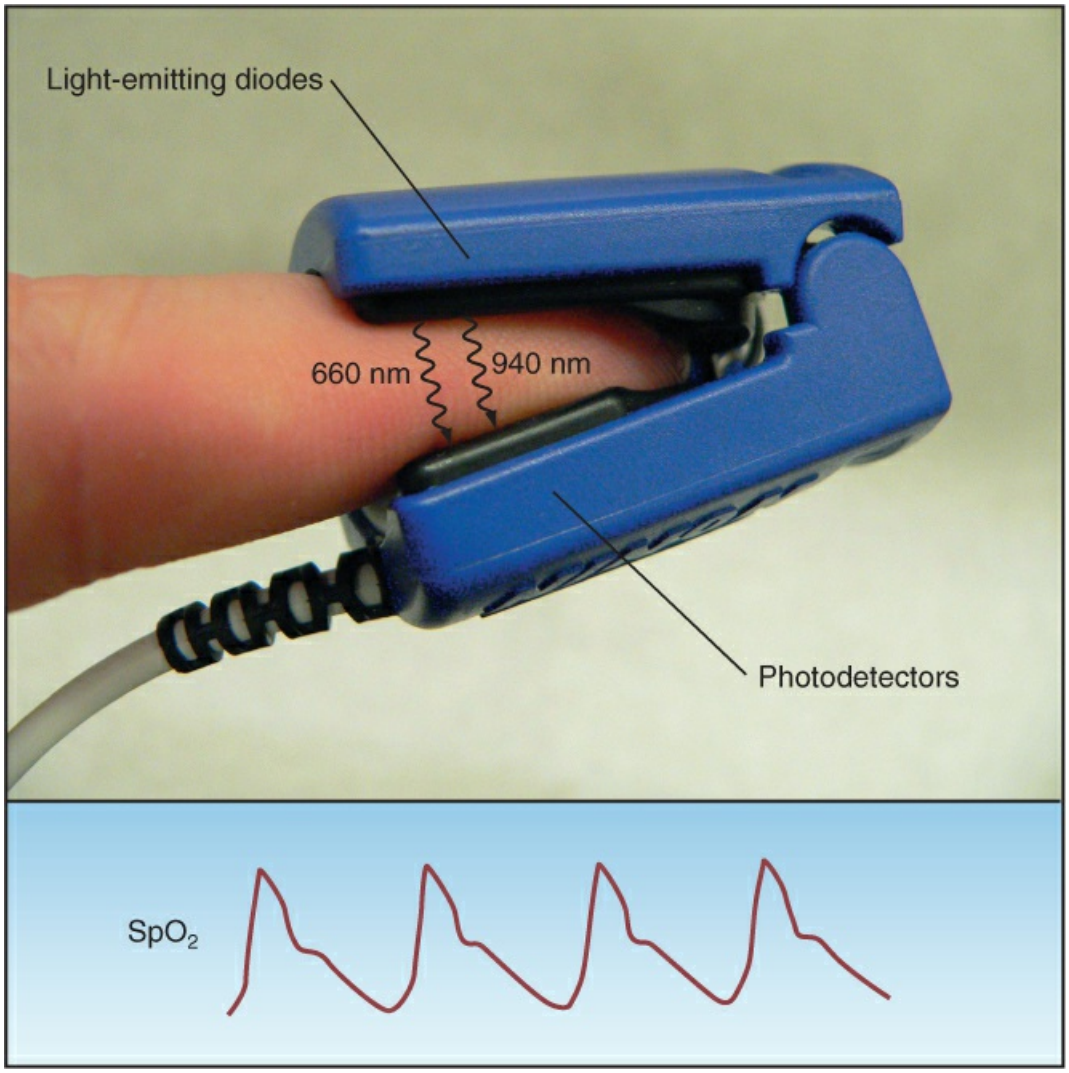


FIGURE 7.3 Pulse oximetry using a fingertip probe. The bottom panel shows the pulsatile photoplethysmography tracing. See text for explanation.

$$SpO_2 = (HbO_2 / Hb + HbO_2) \times 100 \quad (7.1)$$

Forehead Pulse Oximetry

The forehead is an appealing site for pulse oximetry because the arterial circulation in the forehead (which originates from the internal carotid artery) is less prone to vasoconstriction than the digital arteries in the fingers (5). Clinical studies have shown that pulse oximetry in the forehead can provide suitable SpO_2 measurements when fingertip SpO_2 recordings are compromised by hypotension or peripheral vasoconstriction (6).

Unlike fingertip oximetry, which is based on light *transmission*, forehead oximetry relies on light *reflectance*. This is illustrated in the right-sided panel in Figure 7.4. The forehead oximetry probe contains light-emitting diodes and photodetectors positioned next to each other, and the intensity of light reflected back from the underlying arteries is recorded and processed to derive the SpO_2. (*Note:* This is also the method used by smartwatches to monitor arterial oxygenation.)

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THE ELASTIC HEADBAND: Venous pulsations can be prominent in the forehead region (especially during positive-pressure ventilation), and can be misread as arterial pulsations, resulting in spurious SpO_2 recordings. Elastic headbands have been shown to reduce the risk of interference from venous pulsations (7), and these headbands are used routinely with forehead pulse oximetry.

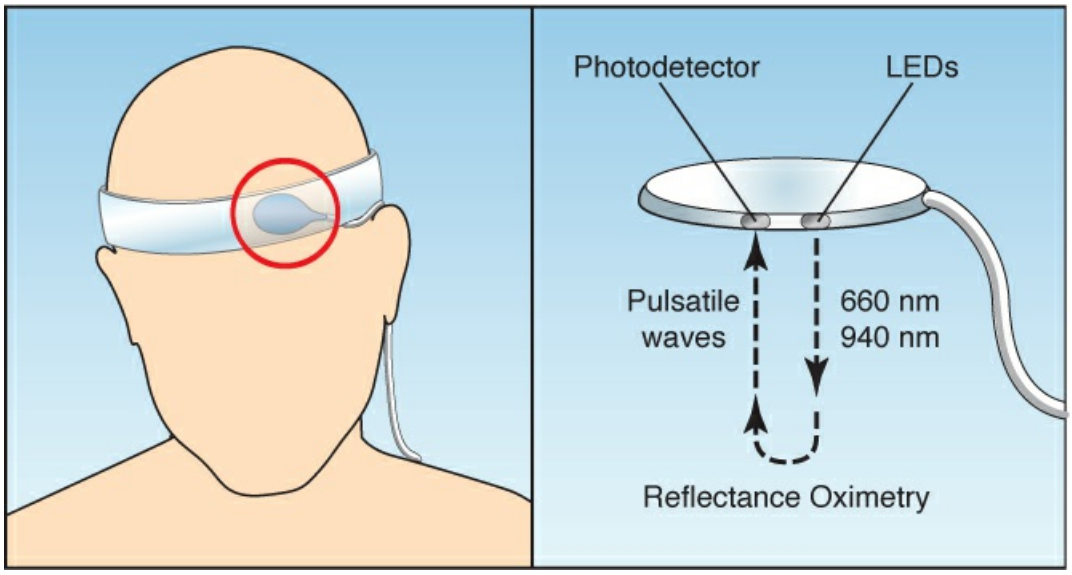


FIGURE 7.4 Forehead pulse oximetry., The forehead probe (circled in red) is placed just above the eyebrows, and is secured with an elastic headband to reduce venous pulsations. This method uses light reflectance to determine the SpO_2. See text for further explanation.

CO-Oximetry

Standard pulse oximeters do not detect carboxyhemoglobin (COHb) or methemoglobin (metHb) in blood. This requires additional wavelengths of light, and these are available in devices called "CO-oximeters" that generate up to 8 wavelengths of light to measure all forms of hemoglobin. These devices are located in clinical laboratories, and require an arterial blood sample. There is also a portable "pulse CO-oximeter" (Rainbow Pulse CO-oximeter, Masimo Corp, Irvine, CA) that has been used by firefighters and emergency personnel to detect carbon monoxide intoxication, but the accuracy of this device has been inconsistent (4).

Normally, metHb and COHb account for less than 5% of the total hemoglobin pool in blood (7,8). When these hemoglobin variants are abnormally elevated, the arterial O_2 saturation (SaO_2) decreases because HbO_2 is a lower fraction of the total hemoglobin pool. However, the SpO_2 from pulse oximetry is only marginally influenced by COHb or metHb levels (9–11). Therefore, *in cases of methemoglobinemia and carbon monoxide poisoning, the SpO_2 overestimates the actual SaO_2, and is not a reliable marker of arterial O_2 desaturation.* When abnormal elevations of metHb or COHb are suspected, an arterial blood sample should be sent to the hospital laboratory for a complete oximetry assessment. (Carbon monoxide poisoning and methemoglobinemia are described in Chapter 53.)

Accuracy

The accuracy of a single SpO_2 measurement is 3–4%, and the accuracy of serial measurements is 2–3% (4). The SpO_2 is usually higher than the arterial O_2 saturation (SaO_2), which may reflect the undetected decrease in SaO_2 from the small amounts of COHb and metHb in blood. The discrepancy between SpO_2 and SaO_2 is greater (by 1–3%) in subjects with dark skin pigmentation (12).

Undetected Hypoxemia

The tendency for the SpO_2 to overestimate the SaO_2 can result in instances of undetected hypoxemia, and these are more frequent in patients with dark skin pigmentation (8,9). In one study that included 88,000 paired measurements of SaO_2 and SpO_2, the incidence of undetected hypoxemia (i.e., SaO_2 <88% with SpO_2 ≥92%) was 6.9% in Blacks, 6.0% in Hispanics, and 4.9% in Whites and Asians. Since these discrepancies were recorded at one point in time, their clinical significance is unclear.

OXIMETRY: APPLICATIONS

Pulse oximetry is indicated in any situation where arterial oxygenation is a concern, and it is considered a requirement for patient safety in certain areas of the hospital (e.g., ICUs and operating suites). Therefore, the important issue concerning pulse oximetry in the ICU is not when to use it, but how to use it. The following information is relevant in this regard.

Arterial Oxygenation

As a surrogate measure of the arterial oxyhemoglobin saturation (SaO_2), the SpO_2 provides information about the oxygen concentration in arterial blood (CaO_2). The determinants of CaO_2 are identified in the following equation:

$$CaO_2 = 1.34 \times [Hb] \times SaO_2 \qquad (7.2)$$

where 1.34 is the oxygen binding capacity of hemoglobin (mL/g), [Hb] is the hemoglobin concentration (g/dL), and SaO_2 is expressed as a percentage, but in decimal notation (e.g., 0.75 instead of 75%). For a healthy adult with [Hb] = 15 g/dL and SaO_2 = 0.98, the CaO_2 is (1.34 × 15 x 0.98) = 19.7 mL/dL (or 197 mL/L).

Target SpO_2

The most recent clinical practice guideline for oxygen therapy recommends a target SpO_2 of 90–94% for most patients, and a slightly lower target (88–92%) for patients with hypercapnic respiratory failure (10). (For more on the target SpO_2 during oxygen therapy, see Chapter 25.) A change in SpO_2 of 3–4% is considered a significant change (4).

Venous Oximetry

Oximetry can also be used to measure the O_2 saturation in venous blood from the pulmonary artery (i.e., mixed venous blood) or the superior vena cava (i.e., central venous blood). This

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measurement can be obtained by sending a blood sample (from either site) to the lab for CO-oximetry, but there are also specialized "oximetry catheters" that can continuously monitor the venous O_2 saturation in the superior vena cava or pulmonary artery. The operation of these catheters is illustrated in Figure 7.5. Oximetry catheters contain fiberoptic bundles that emit two wavelengths of light from the tip of the catheter (the same two wavelength used in pulse oximeters). Another channel of the catheter is connected to a photodetector that records the intensity of light that is reflected back from the hemoglobin in circulating erythrocytes. This technique (reflectance oximetry) is similar to the one used in forehead pulse oximetry. Oximetry catheters process and display the venous O_2 saturation every 5 seconds.

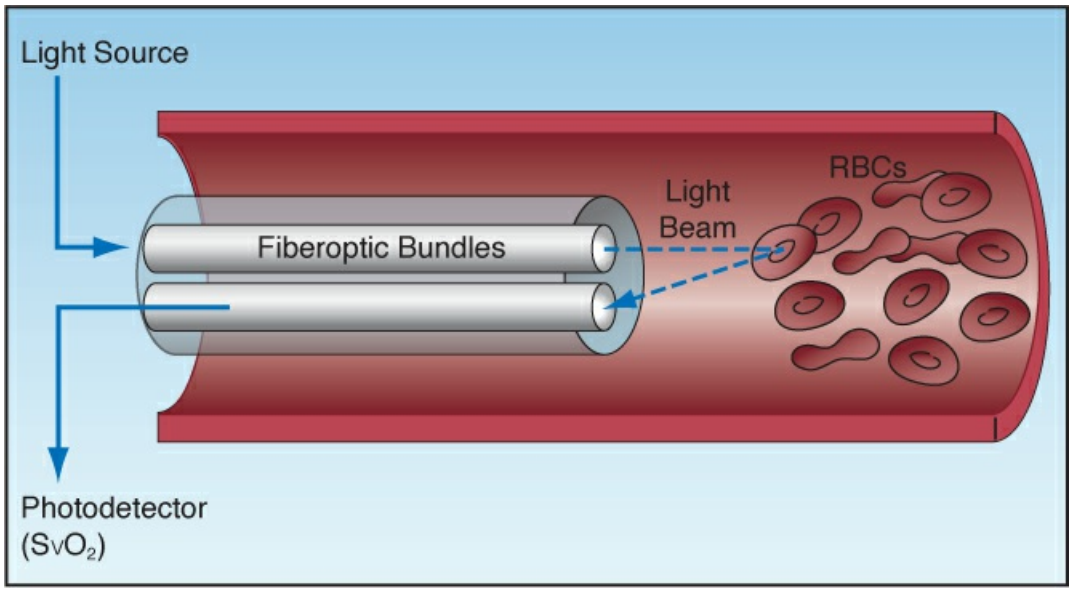


FIGURE 7.5 Continuous monitoring of venous O_2 saturation (SvO_2) using catheters capable of reflectance oximetry. See text for explanation.

Venous O_2 Saturation

The venous O_2 saturation (SvO_2) is a reflection of the balance between systemic O_2 delivery (DO_2) and O_2 consumption (VO_2), as described by the equation below:

$$SvO_2 = 1 - VO_2/DO_2 \tag{7.3}$$

(For the derivation of this equation, see Chapter 9.) Thus, a decrease in SvO_2 from its normal value of 70 – 75% indicates a decrease in O_2 delivery relative to O_2 consumption.

Mixed Venous O_2 Saturation

The O_2 saturation in the pulmonary arteries (called the "mixed venous" O_2 saturation) is an "averaged" SvO_2 from all venous effluents (except the bronchial veins, which drain into the pulmonary veins). This measurement requires a pulmonary artery catheter (described in the next chapter), and there are specialized pulmonary artery oximetry catheters that continuously monitor the SvO_2 with an accuracy of 1–2% (13). Studies in ventilator-dependent patients

showed that the SvO_2 varies an average of 5–6% over just a few hours in patients who are clinically stable (14). Therefore as a general rule, a change in SvO_2 is considered significant if it exceeds 5% and persists for longer than 10 minutes (15).

Central Venous O_2 Saturation

The decreased popularity of the pulmonary artery catheter has shifted attention to the O_2 saturation in the superior vena cava, called the "central venous O_2 saturation" ($ScvO_2$), as a surrogate measure of the SvO_2. The $ScvO_2$ measured by central venous oximetry catheters is slightly lower than the SvO_2, but in patients with circulatory shock, the $ScvO_2$ is greater than the SvO_2 (16). (This change is attributed to the redistribution of blood flow to the cerebral circulation in circulatory shock.) Single measurements of $ScvO_2$ can differ from SvO_2 by as much as 10% in patients who are hemodynamically unstable, but the difference is reduced (to within 5%) when multiple measurements are obtained (17). Therefore, the $ScvO_2$ seems most valuable in identifying *trends* in the balance between DO_2 and VO_2.

Dual Oximetry

The interpretive value of SvO_2 (or $ScvO_2$) can be enhanced by adding the SpO_2 from pulse oximetry. The difference ($SpO_2 – SvO_2$) is roughly equivalent to the O_2 extraction from capillary blood (18), so the following relationships should hold:

$$VO_2 = DO_2 \times (SpO_2 – SvO_2) \tag{7.4}$$

(These relationships are described in detail in Chapter 9.) When O_2 delivery (DO_2) begins to decline (from a decrease in cardiac output, anemia, or hypoxemia), there is an increase in O_2 extraction from capillary blood ($SpO_2 – SvO_2$), and this helps to maintain a constant O_2 consumption (VO_2). The ($SpO_2 – SvO_2$) increases in a linear fashion from a normal value of about 0.25 (25%) up to a value of about 0.50 (50%). Beyond this, any further decrease in DO_2 elicits less of an increase in ($SpO_2 – SvO_2$), and this marks the onset of oxygen-limited metabolism. Therefore, the following interpretations of the ($SpO_2 – SvO_2$) are possible:

. An ($SpO_2 – SvO_2$) >25% indicates that O_2 delivery is reduced (from low cardiac output, anemia, or hypoxemia) relative to O_2 consumption.

. An ($SpO_2 – SvO_2$) >25% but <50% indicates that O_2 delivery is reduced, but not enough to compromise aerobic metabolism.

. An ($SpO_2 – SvO_2$) ≥50% indicates a threat to aerobic metabolism.

The utility of ($SpO_2 – SvO_2$) monitoring is demonstrated by a study showing that an ($SpO_2 – SvO_2$) ≥50% is a valid indicator for the transfusion of packed red blood cells to correct anemia (19).

CAPNOGRAPHY: METHODOLOGY

A L G r a w a n y

The detection of CO_2 in exhaled gas can be done qualitatively (using colorimetric CO_2 detectors to verify entry into the trachea during endotracheal intubation) or quantitatively (using infrared capnography); this section will focus on the latter methodology and how it can be used.

Infrared Capnography

The quantitative method of monitoring exhaled CO_2 is based on the ability of CO_2 to absorb light in the infrared spectrum. The use of this method during mechanical ventilation is illustrated in Figure 7.6 (20,21). An infrared CO_2 probe is placed between the ventilator and the patient, and it emits a continuous infrared light beam that travels through the respired gas (22). On the opposite side of the probe is a rapid-response photodetector that records the intensity of infrared light transmission and processes the information to generate a display of the changes in PCO_2 that occur during each exhalation. This is called a *capnogram*.

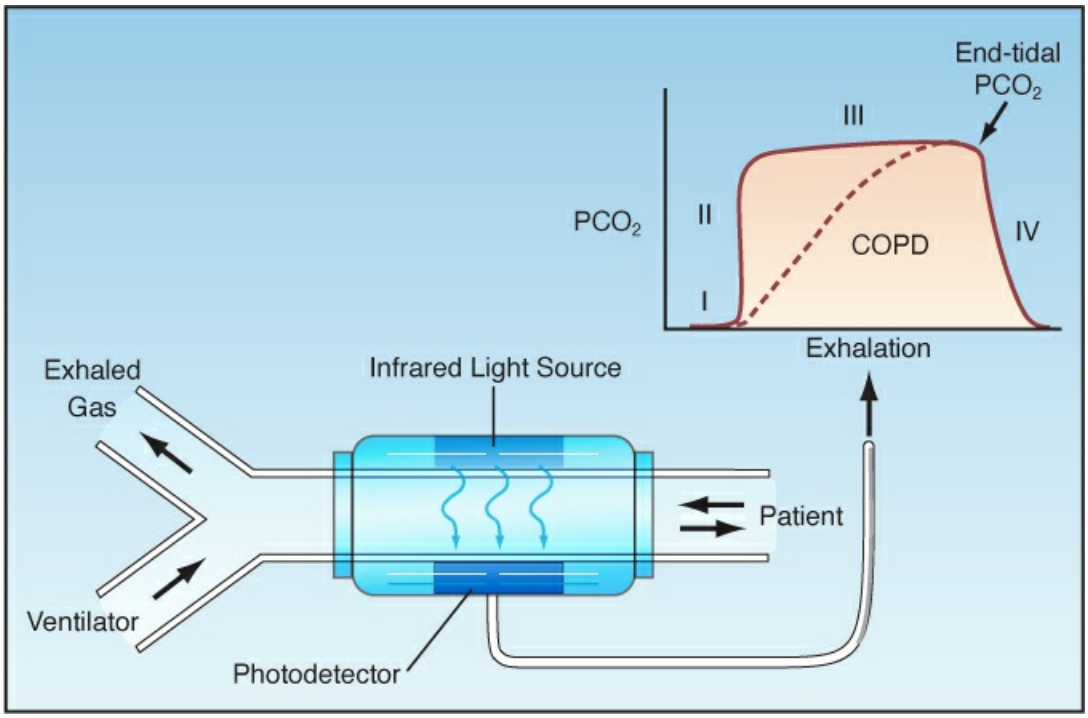


FIGURE 7.6 Infrared capnography during mechanical ventilation. The CO_2 probe sits between the ventilator and the patient, and transmits a continuous infrared light beam through the respired gas. The photodetector processes the transmitted light and generates a record of the changes in PCO_2 that occur during each exhalation (the capnogram). See text for further explanation.

The Capnogram

As shown in Figure 7.6, the capnogram is separated into four phases. Phase I represents the very start of expiration, where the expired gas is from the anatomic dead space, and has negligible amounts of CO_2. In phase II, the exhaled CO_2 steadily rises, and the rate of rise is rapid when alveolar ventilation is well matched to perfusion (i.e., in the normal lung). A slowly rising phase II indicates an increase in alveolar dead space (since these alveoli have a low PCO_2), which occurs in COPD. Phase III is normally a plateau, indicating that alveolar emptying has reached a

steady state, and Phase IV is the onset of inhalation.

Nonintubated Patients

End-tidal CO_2 monitoring is also available for spontaneously breathing (non-intubated) patients using infrared CO_2 probes that are specially designed for nasal cannulas and facemasks. An example of a CO_2 detection system for nasal cannulas is illustrated in Figure 7.7. In this case, a miniaturized infrared CO_2 probe (like the one in Figure 7.6) is placed in-line with a nasal cannula and an adapter that collects orally exhaled gas. Oxygen is delivered through another cannula in the system.

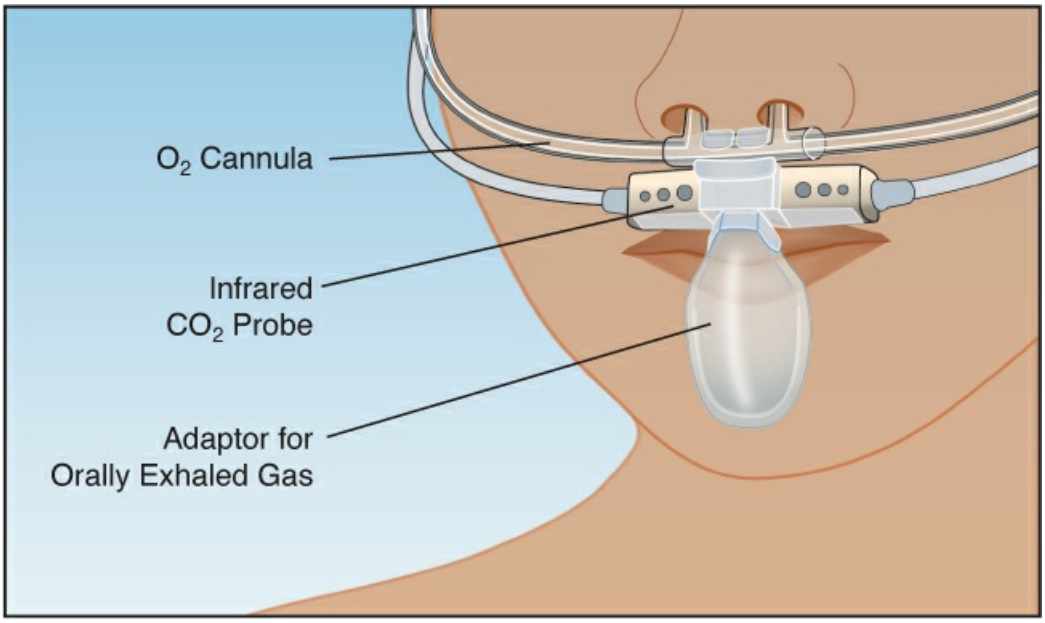


FIGURE 7.7 A modified infrared CO_2 probe (capONE, Nihon Kohden, Tokyo, Japan) for monitoring end-tidal PCO_2 during spontaneous breathing.

CAPNOGRAPHY: APPLICATIONS

Procedures

Procedural Sedation

The sedation used for a variety of procedures creates a risk of hypoventilation, especially in patients with any of the following conditions: advanced age, obesity, sleep apnea, or chronic CO_2 retention from obstructive lung disease. Hypoventilation causes an increase in end-tidal PCO_2 ($P_{ET}CO_2$), and this change occurs before the decrease in SpO_2. Capnography has proven superior to oximetry for detecting hypoventilation during procedural sedation (20,23); as a result, monitoring $P_{ET}CO_2$ is recommended in all instances where procedural sedation is used (24).

Endotracheal Intubation

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For placement of endotracheal tubes, verification of tracheal placement with exhaled CO_2 detection (either with colorimetric or infrared CO_2 detectors) has become a "standard of care" (25) for the following reasons:

. Unrecognized esophageal intubation (which can be fatal) is reported in one of every 18 emergency intubations in critically ill patients (26).

. Clinical assessment, such as auscultation for breath sounds, does not always differentiate between tracheal and esophageal intubation (27).

There are also situations where exhaled CO_2 is negligible despite tracheal placement of an endotracheal tube. This occurs primarily during cardiac arrest, when there is little or no venous return to deliver CO_2 to the alveoli. Severe bronchospasm has also been proposed as a cause of this condition, but this has not been adequately verified (24).

Changes in Cardiac Output

One of the most promising applications of end-tidal PCO_2 monitoring is the detection of acute changes in cardiac output. The correlation between changes in $P_{ET}CO_2$ and changes in cardiac output is demonstrated in Figure 7.8 (28). This relationship suggests that $P_{ET}CO_2$ monitoring might be valuable for detecting cardiac output responses to interventions such as volume loading. In this context, the available studies have shown that changes in $P_{ET}CO_2$ are predictive of fluid responsiveness in mechanically ventilated patients (29), but not in spontaneously breathing healthy patients (30). This application deserves further attention.

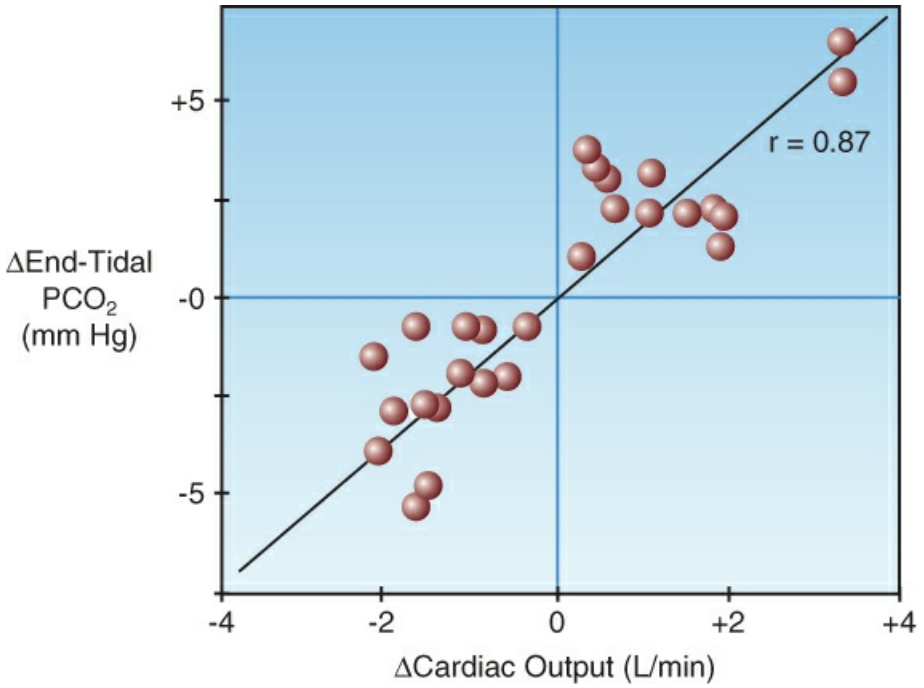


FIGURE 7.8 Relationship between changes in end tidal PCO_2 and changes in cardiac output in a group of postoperative patients. r = correlation coefficient. Data from Reference 28. See text for further explanation.

Cardiopulmonary Resuscitation

One successful application of the relationship between $P_{ET}CO_2$ and cardiac output has been the predictive value of end-tidal CO_2 monitoring during cardiopulmonary resuscitation; i.e., studies indicate that return of spontaneous circulation is unlikely if the $P_{ET}CO_2$ is ≤10 mm Hg after 20 minutes of cardiopulmonary resuscitation (31). This topic is presented in detail in Chapter 21.

$PaCO_2$ – $P_{ET}CO_2$ Difference

Although the $P_{ET}CO_2$ is roughly equivalent to the $PaCO_2$ in healthy subjects, the $P_{ET}CO_2$ will decrease relative to the $PaCO_2$ in any condition associated with an increase in "dead space" ventilation (i.e., when alveoli are overventilated relative to perfusion). Most causes of respiratory failure are accompanied by an increase in physiologic dead space (see the left column in Table 7.1), and the magnitude of the

TABLE 7.1 Conditions that Alter the Relationship Between Arterial and End-Tidal PCO_2

$PETCO_2 < PaCO_2$	$PETCO_2 > PaCO_2$
• Leaky Ventilator Circuit • Excessive Lung Inflation • Pneumonia • Obstructive Lung Disease • Pulmonary Edema • Pulmonary Embolism • Acute Decrease in Cardiac Output	• Hypermetabolism • Metabolic Acidosis • Hyperoxia

($PaCO_2$ - $P_{ET}CO_2$) difference can be used to follow the clinical course of the underlying disorder (21). During mechanical ventilation, an increase in the ($PaCO_2$ - $P_{ET}CO_2$) difference can be partly due to overventilation (e.g., from excessive inflation volumes), so the $PaCO_2$ - $P_{ET}CO_2$ difference can help to identify the optimal inflation volume during mechanical ventilation.

A FINAL WORD

Helping Fill the Void

The fall from grace of the pulmonary artery catheter has created a void in the ability to evaluate tissue perfusion and oxygenation, but oximetry and capnography can help to fill this void in the following ways:

. Monitoring the SpO_2 with pulse oximetry provides no information about tissue oxygenation, but the use of "dual oximetry" to monitor the ($SpO–SvO_2$) difference provides information about the threat of inadequate tissue oxygenation.

. Monitoring the end-tidal PCO_2 ($P_{ET}CO_2$) can be useful for detecting acute changes in cardiac

A L G r a w a n y

output, and it is particularly valuable as a prognostic aid in cardiopulmonary resuscitation.

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A L G r a w a n y

Chapter 8

The Pulmonary Artery Catheter

A searchlight cannot be used effectively without a fairly thorough knowledge of the territory to be searched.

Fergus Macartney (*a*)

In 1970, a vascular catheter was introduced that would revolutionize the practice of bedside hemodynamic monitoring (1). It was a very long, centrally-placed catheter with a small inflatable balloon at its distal end. When inflated, the balloon allowed the flow of venous blood to carry the catheter through the right side of the heart and into one of the pulmonary arteries (like floating down a river on an inflatable inner tube). This balloon flotation effect allowed a right heart catheterization to be performed at the bedside, without the need for fluoroscopic guidance. An added bonus was a temperature-sensitive thermistor on the catheter, which allowed measurements of cardiac output using the *thermodilution* technique. When used appropriately, this *pulmonary artery balloon flotation catheter* could generate a comprehensive set of physiological measurements for evaluating cardiac performance and systemic oxygen transport (2).

Not surprisingly, the pulmonary artery (PA) catheter gained rapid and widespread acceptance, and became a staple of critical care management in the latter years of the twentieth century. However, storm clouds appeared in the early years of the twenty-first century, when a series of (poorly designed) clinical studies showed that PA catheters did not improve survival (3). The fallout from these studies was a swift and precipitous decline in the use of PA catheters; e.g., one large survey showed a 70% decrease in PA catheter use over the years from 1999 to 2013 (4).

The PA catheter's fall from grace is undeserved (see A FINAL WORD at the end of the chapter), as this catheter provides a wealth of information that can be useful if applied appropriately (5,6). The goal of this chapter is to introduce you to these measurements, and how they are obtained. As suggested in the introductory quote, the benefit derived from the PA catheter will depend on your knowledge of cardiovascular physiology.

THE CATHETER

The basic features of a PA catheter are shown in Figure 8.1. The catheter is 110 cm long and has an outside diameter of 2.3 mm (about 7 French). There are two internal channels: one channel emerges at the tip of the catheter (the distal or PA lumen), and the other channel emerges 30 cm proximal to the catheter tip (the proximal or RA lumen), which should be situated in the right atrium. The tip of the catheter has a small inflatable balloon (1.5 mL capacity) that helps to carry the catheter to its final destination. When the balloon is fully inflated, it creates a recess for the tip of the catheter that prevents the tip from scraping along the vessel wall as the catheter is advanced. A small thermistor (a temperature-sensing transducer) is placed near the tip of the catheter, and is used to measure the cardiac output using the thermodilution technique (described later in the chapter).

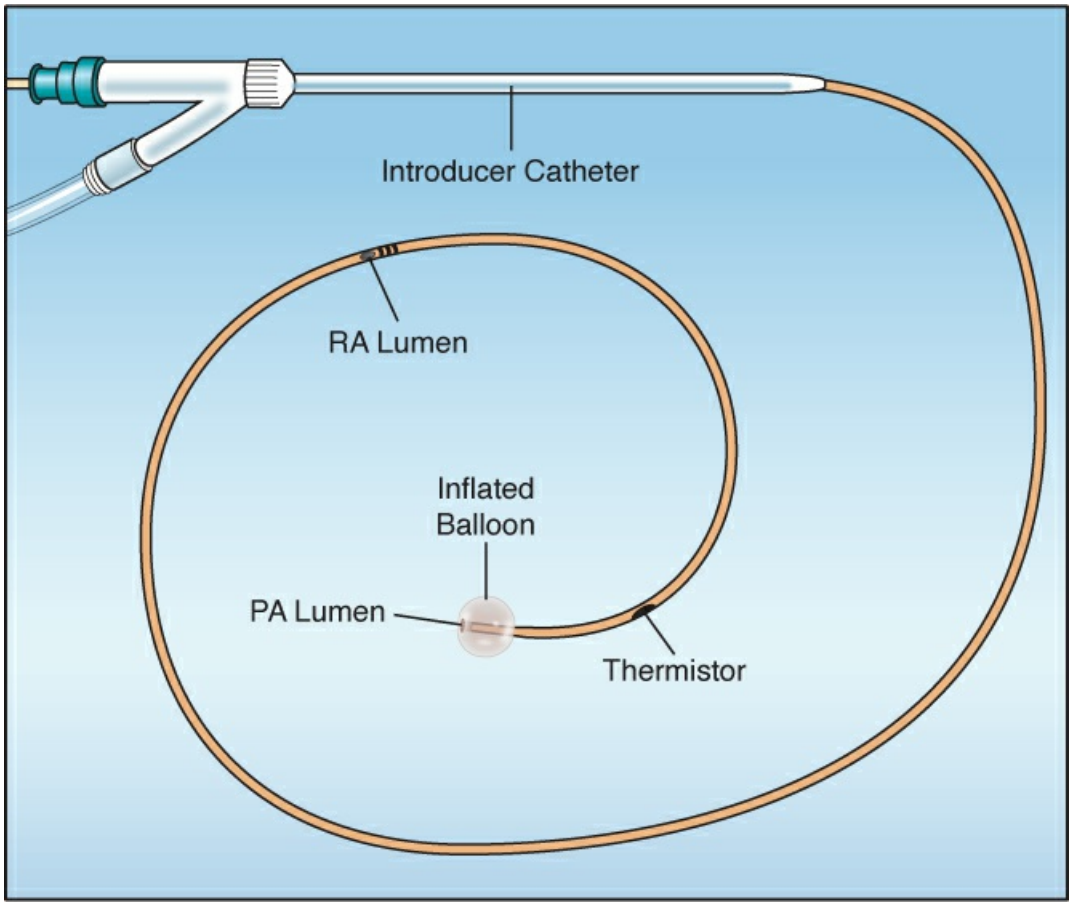


FIGURE 8.1 The basic features of a pulmonary artery (PA) catheter. Note that the PA catheter has been threaded through a large-bore introducer catheter that has a side-arm infusion port.

Catheter Placement

The PA catheter is inserted through a large bore (8–9 French) introducer sheath that has been placed in the subclavian vein or internal jugular vein (see Figure 8.1). The distal lumen of the catheter is attached to a pressure transducer to monitor vascular pressures as the catheter is advanced. When the catheter is passed through the introducer sheath and enters the superior vena cava, a venous pressure waveform appears. When this occurs, the balloon is inflated with 1.5 mL of air, and the catheter is advanced with the balloon inflated. The location of the catheter tip is

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determined by the pressure tracings recorded from the distal lumen, as shown in Figure 8.2.

. The superior vena cava pressure is identified by a venous pressure waveform, which appears as small amplitude oscillations. This pressure remains unchanged until the catheter enters the right ventricle, and it represents the mean pressure in the right atrium, more commonly known as the *central venous pressure*.

. When the catheter tip is advanced across the tricuspid valve and into the right ventricle, a pulsatile waveform appears. The peak (systolic) pressure is a function of the strength of right ventricular contraction, and the lowest (diastolic) pressure is equivalent to the mean right atrial pressure.

. When the catheter moves across the pulmonic valve and into a main pulmonary artery, the pressure waveform shows a sudden rise in diastolic pressure with no change in the systolic pressure. (*Note:* The systolic pressure will increase in the presence of pulmonary hypertension). The rise in diastolic pressure is caused by resistance to flow in the pulmonary circulation.

. As the catheter is advanced along the pulmonary artery, the pulsatile waveform disappears, leaving a nonpulsatile pressure that is normally at the same level as the diastolic pressure of the pulsatile waveform. This is the *pulmonary artery occlusion pressure*, more commonly known as the *wedge pressure*, and is a reflection of the filling pressure on the left side of the heart (see the next section).

. When the wedge pressure tracing appears, the balloon is deflated, and the pulsatile pressure waveform should reappear. The catheter is then secured in place, and the balloon is left deflated. (*Note:* Prolonged balloon inflation creates the risk of pulmonary artery rupture or pulmonary infarction.)

. On occasion, the pulsatile pressure in the pulmonary arteries never disappears despite advancing the catheter maximally. In this situation, the pulmonary artery diastolic pressure can be used as a surrogate measure of the wedge pressure, but only if there is no evidence of pulmonary hypertension (see next section).

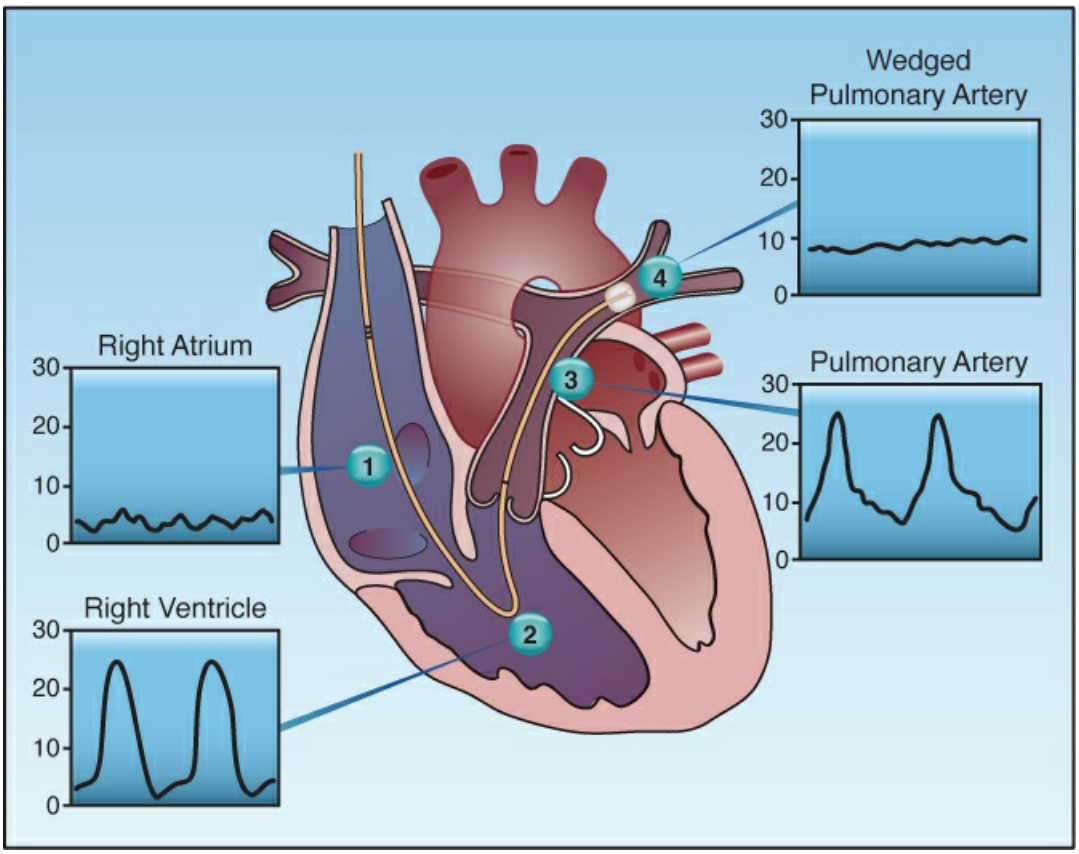


FIGURE 8.2 The pressure waveforms at different points along the normal course of a pulmonary artery catheter. These waveforms are used to identify the location of the catheter tip as it is advanced.

THE WEDGE PRESSURE

The wedge pressure is obtained by slowly inflating the balloon at the tip of the PA catheter until the pulsatile pressure disappears, as shown in Figure 8.3. Note that the wedge pressure is at the same level as the diastolic pressure in the pulmonary artery. This relationship is altered in pulmonary hypertension, where the wedge pressure is lower than the pulmonary artery diastolic pressure.

Wedge Pressure Tracing

The wedge pressure represents the venous pressure on the left side of the heart, and the magnified section of the wedge pressure in Figure 8.3 shows a contour that is similar to the venous pressure on the right side of the heart. The *a* wave is produced by left atrial contraction, the *c* wave is produced by closure of the mitral valve (during isometric contraction of the left ventricle), and the *v* wave is produced by systolic contraction of the left ventricle against a closed mitral valve. These components are often difficult to distinguish, but prominent *v* waves are readily apparent in patients with mitral regurgitation.

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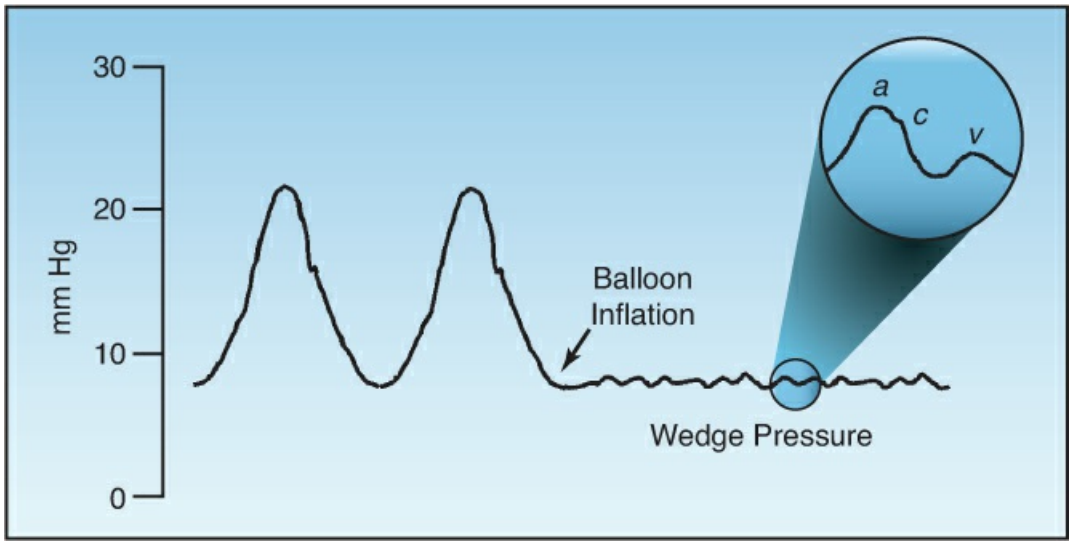


FIGURE 8.3 Pressure tracing showing the transition from a pulsatile pulmonary artery pressure to a balloon occlusion (wedge) pressure. The magnified area shows the components of the wedge pressure: *a* wave (atrial contraction), *c* wave (mitral valve closure), and *v* wave (ventricular contraction).

Wedge Pressure Principle

The underlying principle for the wedge pressure is illustrated in Figure 8.4. When the balloon on the PA catheter is inflated to obstruct flow (Q = 0), there is a static column of blood between the tip of the catheter and the left atrium, and the wedge pressure at the tip of the catheter (P_W) is equivalent to the pulmonary capillary pressure (P_C) and the pressure in the left atrium (P_{LA}). To summarize: if Q = 0, then $P_W = P_C = P_{LA}$. If the mitral valve is behaving normally, the wedge pressure (left atrial pressure) will be equivalent to the left ventricular end-diastolic pressure. Therefore, *in the absence of mitral valve disease, the wedge pressure is a measure of the filling pressure of the left ventricle*.

Respiratory Variations

The wedge pressure is a reflection of the left atrial pressure only if the pulmonary capillary pressure is greater than the alveolar pressure ($P_C > P_A$ in Figure 8.4), otherwise the wedge pressure will reflect the alveolar pressure. This latter condition occurs when there are respiratory variations in the wedge pressure tracing (7). When this occurs, the wedge pressure should be measured at the end of expiration, as explained next.

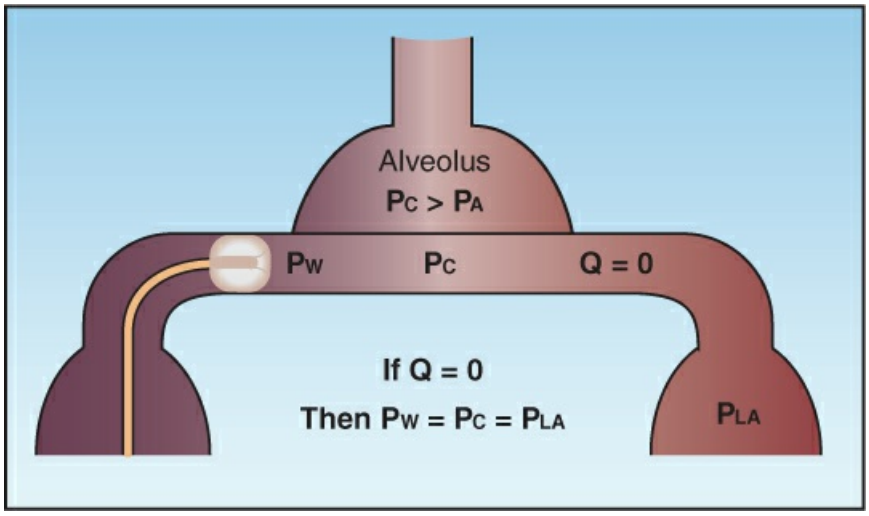


FIGURE 8.4 The basis of the wedge pressure measurement. When flow ceases because of balloon inflation (Q = 0), the wedge pressure (PW) is equivalent to the pulmonary capillary pressure (PC) and the pressure in the left atrium (PLA). This occurs only when the pulmonary capillary pressure is greater than the alveolar pressure (PC > PA).

Changes in intrathoracic pressure can be transmitted into intrathoracic blood vessels (usually low pressure vessels like veins and capillaries), and this creates respiratory variations in the recorded pressure. This is illustrated in Figure 8.5 for a central venous pressure (CVP) tracing. The undulating pattern in the recorded (intravascular) pressure is misleading, because the physiologically important (transmural) pressure is not changing. When this occurs, *the pressure should be measured at the end of expiration, when the intrathoracic pressure is zero (i.e., atmospheric pressure) and the intravascular and transmural pressures are equivalent*. The pressure tracing in Figure 8.5 was recorded during positive pressure mechanical ventilation, so the end-expiratory pressure is the lowest point in each of the pressure undulations. (If this tracing was obtained during spontaneous breathing, the end-expiratory pressure would be the peak pressure for each respiratory cycle.)

POSITIVE END-EXPIRATORY PRESSURE: Positive end-expiratory pressure (PEEP) can falsely elevate the wedge pressure (and CVP) at end-expiration because the intrathoracic pressure is higher than atmospheric (zero) pressure. When PEEP is applied during mechanical ventilation (which is a routine practice), the patient can be briefly disconnected from the ventilator to measure the wedge pressure (8). For patients with ”intrinsic PEEP” (caused by incomplete emptying of the lungs), accurate measurements of the wedge pressure can be difficult (9). Chapter 29 includes a description of intrinsic PEEP and a method for correcting the wedge pressure in the presence of intrinsic PEEP.

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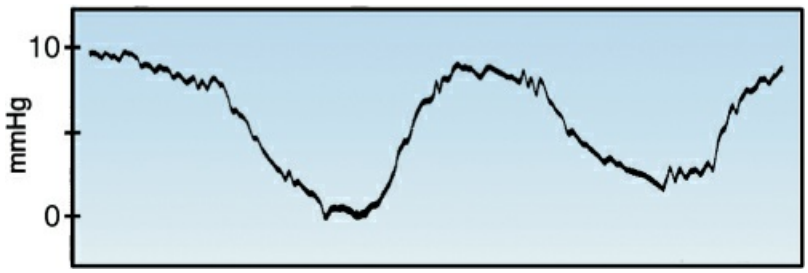


FIGURE 8.5 Respiratory variation in the central venous pressure (CVP) during mechanical ventilation. The CVP should be measured at the end of expiration, which corresponds to the lowest points in the pressure undulations. See text for further explanation.

Spontaneous Variations

In addition to respiratory variations, the wedge pressures can vary spontaneously. This variation in wedge pressure is ≤4 mm Hg in 60% of patients, but it can be as high as 7 mm Hg (10). In general, *a change in the wedge pressure of >4 mm Hg is considered a clinically significant change*.

Wedge vs Hydrostatic Pressure

The wedge pressure is often mistaken as the hydrostatic pressure in the pulmonary capillaries. However, this is not the case (11,12), because the wedge pressure is measured in the absence of blood flow. When the balloon is deflated and flow resumes, the pressure in the pulmonary capillaries (P_C) will be higher than the pressure in the left atrium (P_{LA}), and the difference in pressures will dependent on the flow rate (Q) and the resistance to flow in the pulmonary veins (RV); i.e.,

$$P_C - P_{LA} = Q \times RV \tag{8.1}$$

Since the wedge pressure (P_W) is equivalent to left atrial pressure (P_{LA}), Equation (8.1) can be restated as:

$$P_C - P_W = Q \times RV \tag{8.2}$$

Therefore, the capillary hydrostatic pressure must be greater than the wedge pressure to create a pressure gradient for venous flow to the left side of the heart. The magnitude of this difference is unclear because it is not possible to determine the resistance to flow in the pulmonary veins (RV). However, the $P_C - P_W$ difference will be magnified by conditions that promote pulmonary venoconstriction, such as hypoxemia, endotoxemia, vasopressor infusions, and the acute respiratory distress syndrome (13,14).

Wedge Pressure in ARDS

The wedge pressure has been used to differentiate hydrostatic (cardiogenic) pulmonary edema from the leaky-capillary pulmonary edema that characterizes the acute respiratory distress syndrome (ARDS); i.e., a normal wedge pressure is considered evidence of ARDS (15). However, since the capillary hydrostatic pressure is normally higher than the wedge pressure, and this difference is magnified in ARDS (14), *a normal wedge pressure does not exclude the*

diagnosis of hydrostatic pulmonary edema. Therefore, the use of a normal wedge pressure as a diagnostic criterion for ARDS should be abandoned.

THERMODILUTION CARDIAC OUTPUT

The presence of a temperature-sensitive thermistor on the PA catheter allows a measurement of the cardiac output by the thermodilution technique. This is considered the gold standard method for measuring cardiac output at the bedside.

The Method

The indicator dilution method of measuring blood flow is based on the premise that, when an indicator substance is added to circulating blood, the rate of blood flow is inversely proportional to the change in concentration of the indicator over time. The thermodilution method employs temperature as the indicator.

The thermodilution method is illustrated in Figure 8.6. A bolus (usually 10 mL) of a saline solution that is colder than blood is injected through the proximal port of the catheter in the right atrium. The cold fluid mixes with blood in the right heart chambers, and the cooled blood is ejected into the pulmonary artery and flows past the thermistor on the distal end of the catheter. The thermistor records the change in blood temperature with time. The area under this curve is inversely proportional to flow rate in the pulmonary artery, which is equivalent to the cardiac output (in the absence of intracardiac shunts). Electronic PA monitors integrate the area under the temperature–time curves and provide a digital display of the cardiac output.

Thermodilution Curves

Examples of thermodilution curves are shown in Figure 8.7. The low cardiac output curve (upper panel) has a gradual rise and fall, whereas the high output curve (middle panel) has a rapid rise, an abbreviated peak, and a steep downslope. Note that the area under the low cardiac output curve is greater than the area under the high output curve (i.e., indicating that the area under these curves is inversely related to the flow rate).

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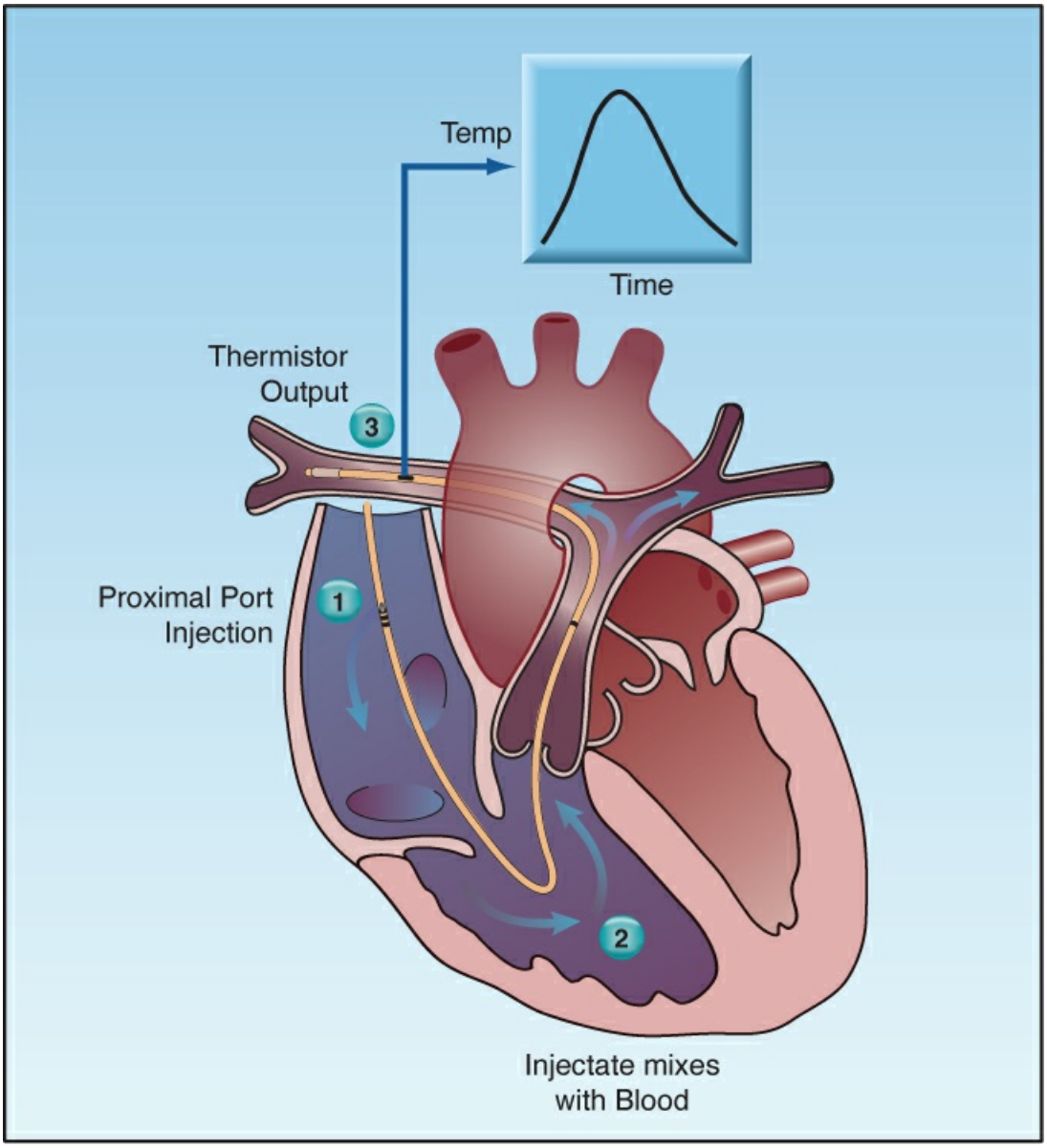


FIGURE 8.6 The thermodilution method of measuring cardiac output. See text for explanation.

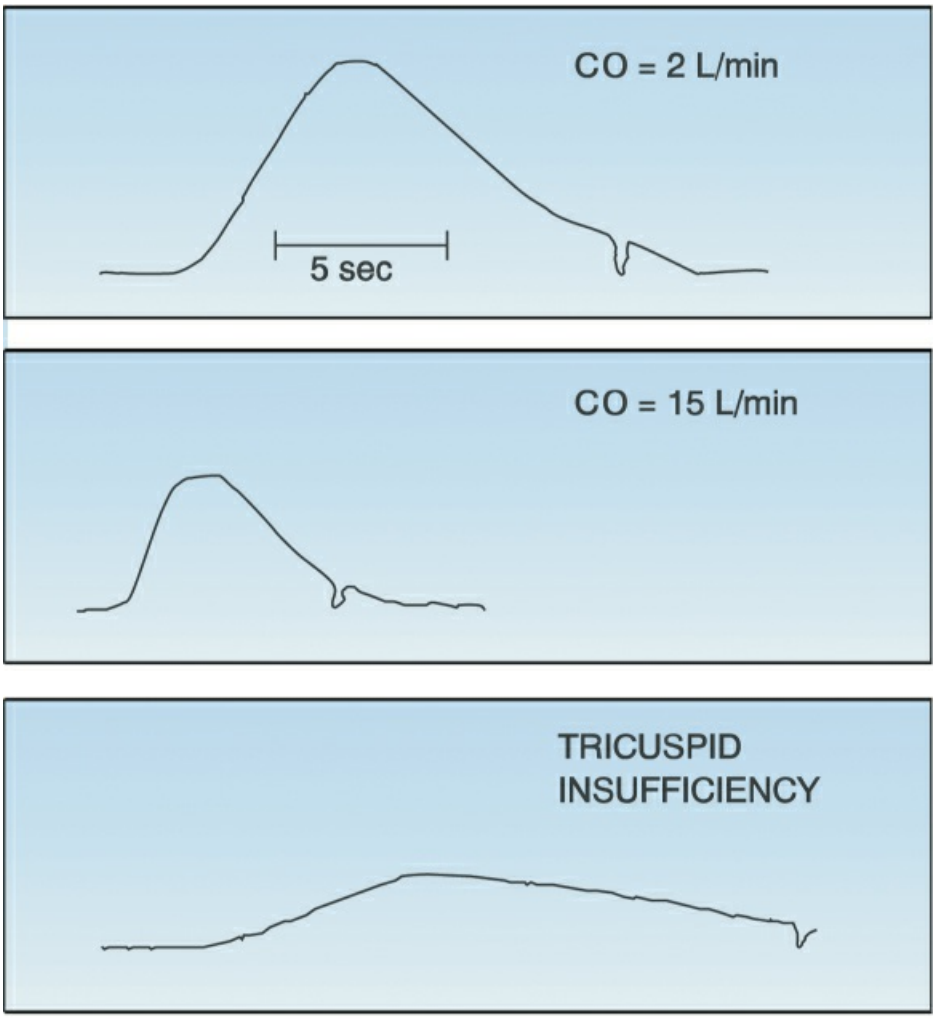


FIGURE 8.7 Thermodilution curves for a low cardiac output (upper panel), a high cardiac output (middle panel), and tricuspid insufficiency (lower panel). The sharp inflection in each curve marks the end of the measurement period. CO = cardiac output.

How Many Measurements?

Serial measurements are recommended for each cardiac output determination. Three measurements are sufficient if they differ by 10% or less, and the cardiac output is taken as the average of all measurements. Serial measurements that differ by more than 10% are considered unreliable (16).

Variability

Thermodilution cardiac output can vary by as much as 10% without any apparent change in the clinical condition of the patient (17). Therefore, *a change in thermodilution cardiac output should exceed 10% to be considered clinically significant.*

Sources of Error

As mentioned earlier, thermodilution is considered the gold standard method for measuring cardiac output at the bedside. However there are sources of error in the methodology, and the

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more notable ones are summarized here.

Operator Error

Low-volume or inappropriately warm injectates can result in overestimation of the cardiac output, and variability in the timing of the injectate in relation to the respiratory cycle can also produce spurious results (6).

Tricuspid Regurgitation

Tricuspid regurgitation can result in underestimation or overestimation of the cardiac output (6). Regurgitant flow across the tricuspid valve causes the injectate to be recycled, producing a prolonged, low amplitude thermodilution curve similar to the one in the bottom frame of Figure 8.7. This results in a falsely low cardiac output measurement (18). Overestimation of the cardiac output can occur when the injectate is warmed because of recycling across the tricuspid valve.

Intracardiac Shunts

Intracardiac shunts produce falsely elevated cardiac output measurements. In right-to-left shunts, a portion of the cold indicator fluid passes through the shunt, thereby creating an abbreviated thermodilution curve similar to the high-output curve in the middle panel of Figure 8.7. In left-to-right shunts, the thermodilution curve is abbreviated because the shunted blood increases the blood volume in the right heart chambers, and this dilutes the indicator solution that is injected.

Continuous Cardiac Output Measurements

Pulmonary artery catheters are available that can provide thermodilution cardiac output measurements as rapidly as every 20 seconds, without the need for bolus injections of cold fluid. These catheters have a small heating filament (near the proximal port of the catheter) that is activated by brief, randomly-generated electrical pulses that raise the surrounding blood temperature by about 0.02°C. The downstream thermistor averages the temperature-time curves over a 3 minute period to determine the cardiac output, which is updated every 20–30 seconds. (The small temperature change creates a poor signal-to-noise ratio, which is why multiple temperature-time curves must be measured and averaged.) This “pulsed warm thermodilution technique” has shown a good correlation with the traditional “intermittent bolus” technique (19,20).

The automated continuous thermodilution technique provides a more dynamic assessment of the cardiac output, and it also eliminates the time commitment, risk of operator error, and the volume infusions associated with the intermittent bolus technique. Moreover, there are PA catheters that allow continuous monitoring of both cardiac output and mixed venous O_2 saturation (SvO_2) (21), which adds the ability to monitor an indirect measure of tissue oxygenation. (See Chapter 9 for an explanation of the SvO_2.)

CATHETER-DERIVED HEMODYNAMICS

The PA catheter provides a wealth of information on cardiovascular function and systemic oxygen transport The following is a description of 10 hemodynamic parameters that can be

monitored using PA catheters. These are listed in Table 8.1.

Body Size

Hemodynamic parameters are often expressed in relation to body size, and the favored measure of body size for hemodynamic measurements is the body surface area (BSA), which can be determined with the following simple equation (22):

$$\text{BSA } (m^2) = \text{Ht (cm)} + \text{Wt (kg)} - 60/100 \quad (8.3)$$

(*Note:* BSA is preferred over weight as an index of body size because cardiac output is linked to metabolic rate, and the basal metabolic rate is expressed in terms of body surface area.) Measurements that are expressed in relation to BSA are referred to as an "index" (e.g., cardiac index instead of cardiac output). The average-sized adult has a body surface area of 1.7 m^2.

TABLE 8.1 Measurements Available with PA Catheters

Parameter	Abbreviation	Normal Range
Central Venous Pressure	CVP	0–5 mm Hg
Pulmonary Artery Wedge Pressure	PAWP	6–12 mm Hg
Cardiac Index	CI	2.4–4.0 L/min/m^2
Stroke Index	SI	20–40 mL/m^2
Systemic Vascular Resistance Index	SVRI	25–30 Wood Units†
Pulmonary Vascular Resistance Index	PVRI	1–2 Wood Units†
Mixed Venous O_2	SvO_2	70–75%
Oxygen Delivery (Index)	DO_2	520–570 mL/min/m^2
Oxygen Uptake (Index)	VO_2	110–160 mL/min/m^2
Oxygen Extraction Ratio	O_2ER	0.2–0.3

†mm Hg /L/min/m^2.

Cardiovascular Parameters

The following parameters are used to evaluate cardiac performance and global tissue perfusion. The reference ranges for these parameters are included in Table 8.1 Size-adjusted parameters are expressed as an *index* (e.g., cardiac index instead of cardiac output).

Central Venous Pressure

When the PA catheter is properly placed, the proximal port of the catheter will be in the right atrium, and the pressure recorded from this port should be the right atrial pressure (PRA). As mentioned previously, the pressure in the right atrium is the same as the pressure in the superior vena cava, and these pressures are collectively called the central venous pressure (CVP). In the absence of tricuspid valve dysfunction, the CVP should be equivalent to the right ventricular end diastolic pressure (RVEDP).

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$$CVP = P_{RA} = RVEDP \tag{8.4}$$

The CVP is then a measure of the filling pressure for the right side of the heart. The normal range for the CVP is 0–5 mm Hg, and it can be a negative pressure in the sitting position.

Wedge Pressure

As described earlier, the pulmonary artery wedge pressure (PAWP) is a measure of the left atrial pressure (P_{LA}), and this pressure should be equivalent to the left ventricular end diastolic pressure (LVEDP) when mitral valve function is normal.

$$PAWP = P_{LA} = LVEDP \tag{8.5}$$

The wedge pressure is thus a measure of the filling pressure for the left side of the heart. This pressure is slightly higher than the CVP (to keep the foramen ovale closed), and the normal range is 6–12 mm Hg.

Cardiac Index

The cardiac output (CO) measurement is the single most important contribution of the PA catheter to hemodynamic assessments, as it not only provides an assessment of cardiac contractile strength (using the stroke volume), it also creates the ability to monitor systemic oxygen transport (see later). The cardiac output is typically expressed in relation to body surface area (BSA), and is called the *cardiac index* (CI).

$$CI = CO/BSA \tag{8.6}$$

The normal range for the cardiac index is 2.4–4 L/min/m^2, which is typically about 60% of the normal range for the cardiac output.

Stroke Index

The stroke volume (i.e., the volume of blood ejected during systole) is used to assess the strength of ventricular contractions, and the relationship between stroke volume and the pulmonary artery wedge pressure can be used to generate ventricular function curves. When adjusted for body size, the stroke volume is called the *stroke index* (SI), and it is derived as the cardiac index (CI) divided by the heart rate (HR):

$$SI = CI/HR \tag{8.7}$$

The normal range of the SI in adults is 20–40 mL/m^2.

Systemic Vascular Resistance Index

The hydraulic resistance in the systemic circulation is not a measurable quantity for a variety of reasons (e.g., resistance is flow-dependent and varies in different regions). Instead, the systemic vascular resistance (SVR) is a global measure of the relationship between systemic pressure and flow. The SVR is directly related to the mean pressure drop from the aorta to the right atrium (i.e., mean arterial pressure or MAP minus the CVP), and is inversely related to the cardiac

output. The size-adjusted SVR index (SVRI) is derived using the cardiac index (CI).

$$SVRI = (MAP - CVP)/CI \tag{8.8}$$

The SVRI is expressed in Wood units (mm Hg/L/min/m^2), which can be multiplied by 80 to obtain more conventional units of resistance (dynes·sec^{-1}·cm^{-5}/m^2), but this conversion offers no advantage (23).

Pulmonary Vascular Resistance Index

The pulmonary vascular resistance (PVR) has the same limitations as mentioned for the systemic vascular resistance. The PVR is a global measure of the relationship between pressure and flow in the lungs, and is derived as the mean pressure drop from the pulmonary artery to the left atrium, divided by the cardiac output. Because the pulmonary artery wedge pressure (PAWP) is equivalent to the left atrial pressure, the pressure gradient across the lungs can be expressed as the difference between the mean pulmonary artery pressure and the wedge pressure (PAP – PAWP). The size-adjusted PVR index (PVRI) is derived using the cardiac index (CI):

$$PVRI = (PAP - PAWP)/CI \tag{8.9}$$

Like the SVRI, the PVRI is expressed in Wood units (mm Hg/L/min/m^2).

Oxygen Transport Parameters

The oxygen transport parameters provide a global measure of oxygen supply and oxygen consumption. These parameters are described in detail in Chapter 9.

Oxygen Delivery

The rate of oxygen transport in arterial blood is called the *oxygen delivery* (DO_2), and is the product of the cardiac output and the oxygen concentration in arterial blood (CaO_2).

$$DO_2 = CO \times CaO_2 \tag{8.10}$$

The O_2 concentration in arterial blood (CaO_2) is a function of the hemoglobin concentration (Hb), the O_2 carrying capacity of hemoglobin (1.34 mL per gram), and the arterial oxyhemoglobin saturation (SaO_2): i.e., $CaO_2 = 1.34 \times Hb \times SaO_2$. Therefore, the DO_2 equation can be rewritten as:

$$DO_2 = CO \times (1.34 \times Hb \times SaO_2) \tag{8.11}$$

The size-adjusted DO_2 is calculated using the cardiac index instead of cardiac output, and it is expressed as mL/min/m^2. The normal range is shown in Table 8.1

Oxygen Uptake

Oxygen uptake (VO_2), also called *oxygen consumption*, is the rate at which oxygen is taken up from the systemic capillaries into the tissues. The VO_2 is calculated as the product of the cardiac

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output (CO) and the difference in O_2 concentration between arterial and venous blood ($CaO_2 - CvO_2$). The venous blood in this instance is "mixed" venous blood in the pulmonary artery.

$$VO_2 = CO \times (CaO_2 - CvO_2) \tag{8.12}$$

If the CaO_2 and CvO_2 are each broken down into their component parts, the VO_2 equation can be rewritten as:

$$VO_2 = CO \times 1.34 \times Hb \times (SaO_2 - SvO_2) \tag{8.13}$$

SaO_2 and SvO_2 are the oxyhemoglobin saturations in arterial and mixed venous blood, respectively. The SaO_2 is continuously monitored with pulse oximetry, and the SvO_2 can also be continuously monitored with specialized PA catheters (mentioned earlier). The SvO_2 can also be measured in a blood sample drawn through the distal port of the PA catheter.

The size-adjusted VO_2 is calculated using the cardiac index instead of the cardiac output, and it is expressed as mL/min/m^2. The normal range is shown in Table 8.1. Note that the VO_2 is much lower than the DO_2 (see next).

Oxygen Extraction Ratio

The oxygen extraction ratio (O_2 ER) is the ratio of O_2 uptake to O_2 delivery:

$$O_2 \text{ ER} = VO_2/DO_2 \tag{8.14}$$

The normal range for the O_2 ER is 0.2 to 0.3, which means that O_2 uptake into tissues represents only 20% to 30% of the O_2 delivered to the systemic microcirculation. Oxygen extraction can increase to 0.50 or even higher to compensate for decreases in O_2 delivery. (See Chapter 9 for a more detailed explanation of this topic.)

Applications

The following are some examples of clinical situations where the information generated by PA catheters can be useful:

. The management of acute, decompensated heart failure, especially when complicated by labile blood pressures or worsening renal function.

. Perioperative management of patients undergoing cardiac surgery.

. The management of circulatory shock (24), especially cardiogenic shock, where use of the PA catheter has had some success in improving outcomes (25).

. Identifying responsiveness to volume infusion (6).

. Determining the need for, and response to, erythrocyte transfusions (see Chapter 12). Ditto for oxygen therapy (see Chapter 25).

. Identifying occult cases of weaning-induced cardiac dysfunction (26) in patients who have difficulty weaning from mechanical ventilation.

The measurements generated by PA catheters will appear in many chapters in this book, including Chapter 9 (Systemic Oxygenation), Chapter 11 (Fluid Management), Chapter 12 (Anemia and Red Blood Cell Transfusions), Chapters 14–17 (Shock Syndromes), Chapter 18 (Acute Heart Failure), Chapter 25 (Oxygen Inhalation), Chapter 26 (Noninvasive Ventilation), and Chapter 30 (Discontinuing Mechanical Ventilation).

A FINAL WORD

An Undeserved Reputation

Despite the wealth of physiologically relevant information it provides, the PA catheter has been vilified and almost abandoned because clinical studies have shown no survival benefit associated with use of the catheter (3). The following are some points in support of the PA catheter.

. First and foremost, *the PA catheter is a monitoring device, not a therapy*, and clinical outcomes (e.g., mortality rate) should be used to evaluate therapies, not measurements.

. Surveys indicate that *physicians often don't understand the measurements provided by PA catheters* (27,28), and any tool can be ineffective in the wrong hands.

. Finally, the incessant use of mortality rates to evaluate critical care interventions is problematic because *the presumption that every intervention has to save lives to be of value is not valid*. Interventions have more specific and immediate goals other than life or death (e.g., the goal of a vasopressor drug infusion is to increase the blood pressure). In the case of a monitoring device, the goal is to provide useful information, and the PA catheter achieves this goal with distinction.

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Chapter 9

Systemic Oxygenation

If one were to name the universal factor in all death…
it would certainly be loss of oxygen.

Sherwin Nuland, MD (*a*)

The management of critically ill patients is dominated by interventions that promote tissue oxygenation, yet it is not possible to monitor tissue O_2 levels in a clinical setting. Instead, tissue oxygenation is evaluated indirectly using global measures of systemic oxygen transport (i.e., O_2 delivery and O_2 uptake) along with a presumed marker of inadequate tissue oxygenation (i.e., the plasma lactate level). This chapter describes these measures, and how they are used (and misused). The information in this chapter is part of the core knowledge base in critical care; i.e., the knowledge that distinguishes ICU-level care from patient care on a general medical or surgical service.

OXYGEN IN BLOOD

The relevant measures of oxygen (O_2) in blood include the partial pressure of O_2 (PO_2), the O_2 saturation of hemoglobin (SO_2), the concentrations of hemoglobin-bound O_2 and dissolved O_2, and the total O_2 concentration (also called the O_2 content). The reference values for these measures in arterial and venous blood are shown in Table 9.1.

TABLE 9.1 Normal Measures of Oxygen in Arterial and Venous Blood

Measure	Arterial Blood	Venous Blood
Partial Pressure of O_2	90 mm Hg	40 mm Hg
O_2 Saturation of Hb	98%	73%
Hb-bound O_2	19.7 mL/dL	14.7 mL/dL
Dissolved O_2	0.27 mL/dL	0.1 mL/dL

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Total O_2 Content[1]	20 mL/dL	15 mL/dL
Blood Volume[2]	1.25 L	3.75 L
Total Volume of O_2	250 mL	555 mL

Values shown are for a body temperature of 37 °C and a hemoglobin concentration of 15 g/dL.

[1]Numbers rounded to nearest whole number.

[2]Volume estimates based on a total blood volume of 5 Liters, with 25% of the volume in arteries and 75% in veins.

Oxyhemoglobin Dissociation Curve

The oxygenation of the hemoglobin pool is expressed as the *oxygen saturation of hemoglobin* (SO_2), and is the fraction of the hemoglobin pool that is fully saturated with oxygen. The SO_2 is measured (by oximetry) as the ratio of oxygenated hemoglobin to total hemoglobin. (See Chapter 7 for a description of oximetry.) The principal determinants of SO_2 are the PO_2 in blood, and the affinity of hemoglobin for binding O_2. These relationships are described by the oxyhemoglobin dissociation curve, which is shown in Figure 9.1. The "S" shape of the curve offers two advantages. First, the arterial PO_2 (PaO_2) is normally on the upper, flat part of the curve, which means that a large drop in PaO_2 (down to 60 mm Hg) results in only minor changes in the oxygenation of hemoglobin (the SO_2). Secondly, the venous PO_2 (which is ≈40 mm Hg) is on the steep portion of the curve, which facilitates O_2 uptake in the pulmonary capillaries.

SHIFTS IN THE CURVE: A number of conditions can alter the affinity of hemoglobin for O_2 and thereby shift the position of the oxyhemoglobin dissociation curve. These conditions are listed in Figure 9.1. A shift of the curve to the right facilitates O_2 release in tissues, while a shift to the left impedes O_2 release. The position of the curve is indicated by the P50, which is the PO_2 that corresponds to an SO_2 of 50%. The P50 is normally about 27 mm Hg (1), and will increase when the curve shifts to the right, and decrease when the curve shifts to the left. For example, a decrease in the P50 to 15 mm Hg (indicating a leftward shift of the curve) has been reported in blood that is stored in acid-citrate-dextrose (ACD) preservative for 3 weeks, and is attributed to depletion of 2,3 diphosphoglycerate (2,3-DPG) in the red blood cells (2).

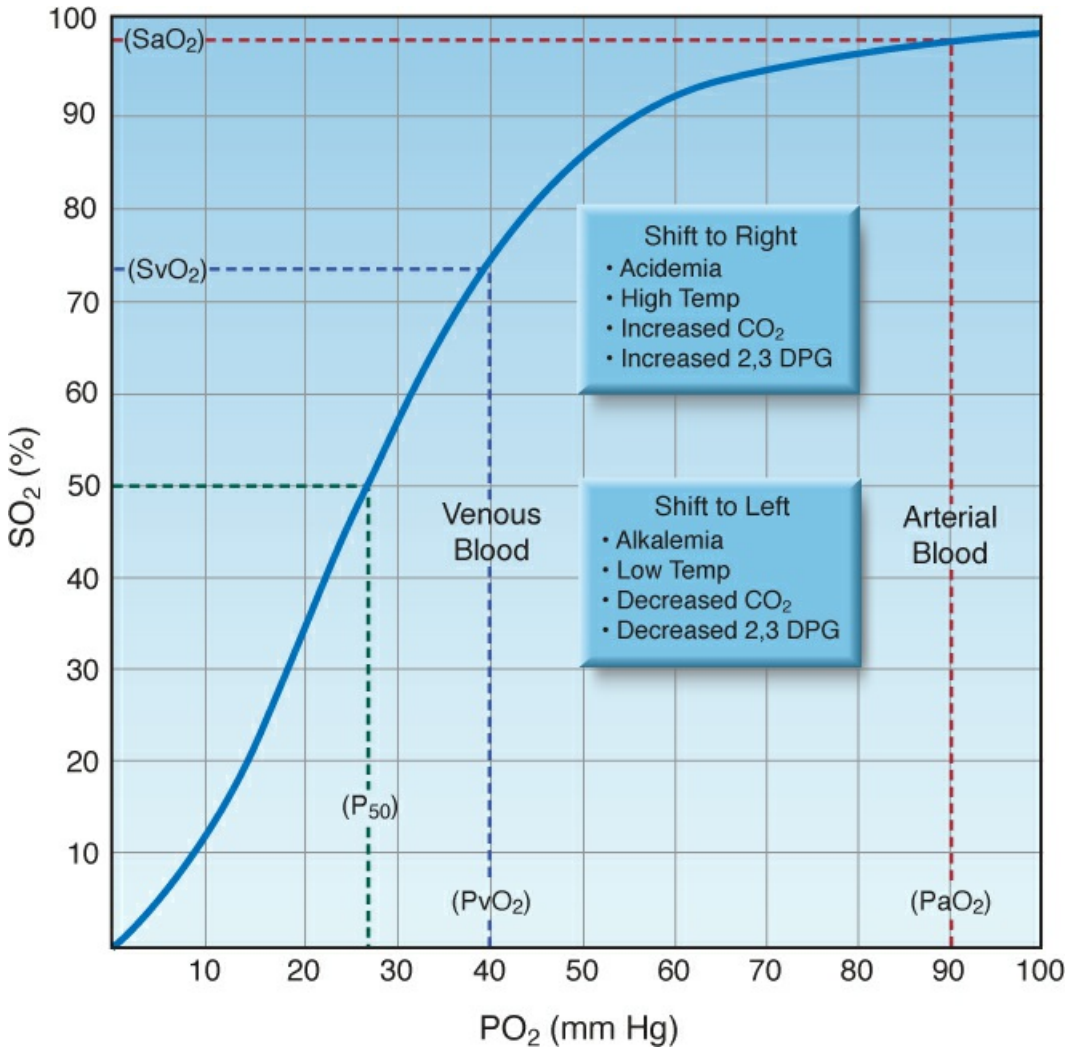


FIGURE 9.1 Oxyhemoglobin dissociation curve showing the normal relationship between the PO_2 in blood and the O_2 saturation of hemoglobin (SO_2). The P_{50} is the PO_2 that corresponds to an SO_2 of 50%. Abbreviations: PaO_2 = arterial PO_2, PvO_2 = venous PO_2, SaO_2 = arterial SO_2, SvO_2 = venous SO_2.

Shifts in the oxyhemoglobin dissociation curve have opposing effects in the pulmonary and systemic capillaries; i.e., a shift to the left impedes O_2 release systemically but facilitates O_2 uptake in the lungs. Therefore, the significance of these shifts for overall O_2 exchange is unclear.

Oxygen Content of Blood

The concentration (content) of O_2 in blood is the summed contribution of the O_2 that is bound to hemoglobin and the O_2 that is dissolved in plasma.

Hemoglobin-Bound Oxygen

The concentration of hemoglobin-bound O_2 (HbO_2) is determined by the concentration of hemoglobin [Hb] (usually expressed as grams per deciliter (100 mL), the O_2 binding capacity of hemoglobin (which is 1.34 mL O_2 per gram Hb), and the O_2 saturation of hemoglobin (SO_2).

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This relationship is expressed as follows (3):

$$HbO_2 = 1.34 \times [Hb] \times SO_2 \text{ (mL/dL)} \quad (9.1)$$

(The SO_2 in this equation is expressed as a decimal rather than percentage: e.g., 0.75 instead of 75%.)

Dissolved Oxygen

Plasma is about 93% water, and oxygen does not readily dissolve in water (which is why hemoglobin is needed as an O_2 carrier). The solubility of O_2 in plasma is temperature-dependent, and varies inversely with a change in body temperature. At a normal body temperature (37 °C), each increment in PO_2 of 1 mm Hg will increase the concentration of dissolved O_2 in plasma by 0.03 mL/L (4). This is expressed as a *solubility coefficient* of 0.03 mL/L/mm Hg (or 0.003 mL/dL/mm Hg) The concentration of dissolved O_2 in plasma at 37 °C is then determined as follows:

$$\text{Dissolved } O_2 = 0.003 \times PO_2 \text{ (mL/dL)} \quad (9.2)$$

This equation emphasizes the limited volume of dissolved O_2 in the physiological PO_2 range (see next).

Arterial O_2 Content

The O_2 content in arterial blood (CaO_2) is determined by combining Equations 9.1 and 9.2 and inserting the SO_2 and PO_2 of arterial blood (SaO_2 and PaO_2): i.e.,

$$CaO_2 = (1.34 \times [Hb] \times SaO_2) + (0.003 \times PaO_2) \text{ (mL/dL)} \quad (9.3)$$

Using normal values for {Hb] (15 g/dL), SaO_2 (0.98), and PaO_2 (90 mm Hg) yields the following:

$$CaO_2 = (19.7) + (0.27) \quad (9.4)$$

Thus, the normal arterial O_2 content is about 20 mL/dL (or 200 mL/L), and *only 1.5% represents dissolved oxygen*.

Anemia vs. Hypoxemia

The relative impact of anemia and hypoxemia on the arterial O_2 content (CaO_2) is shown in Figure 9.2. Note that a 50% reduction in [Hb] (from 15 to 7.5 g/dL) results in an equivalent 50% reduction in CaO_2 (from 20 to 10 mL/dL), while a 50% reduction in the PaO_2 (from 90 to 45 mm Hg, corresponding to a decrease in SaO_2 from 98% to 78%) results in only a 20% decrease in CaO_2 (from 20 to 16 mL/dL). This demonstrates that *anemia has a much greater influence on arterial oxygenation than hypoxemia*.

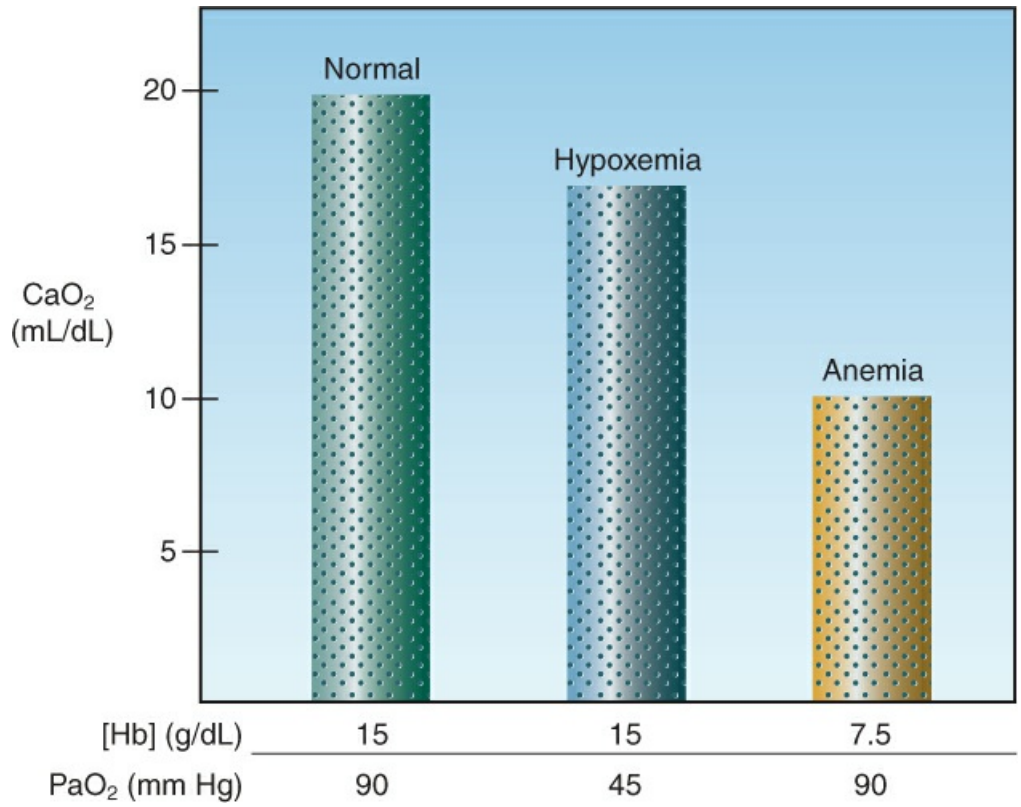


FIGURE 9.2 The influence of proportional degrees of anemia and hypoxemia on the oxygen content in arterial blood (CaO_2).

Venous O_2 Content

The venous O_2 content (CvO_2) represents the O_2 content in "mixed" venous blood (from the right heart or pulmonary artery). The equation describing CvO_2 is similar in format to Equation 9.3, but the SO_2 and PO_2 are for mixed venous blood (SvO_2 and PvO_2).

$$CvO_2 = (1.34 \times [Hb] \times SvO_2) + (0.003 \times PvO_2) \quad (9.5)$$

As shown in Table 9.1, the normal mixed venous O_2 content is about 15 mL/dL, and less than 1% (0.1 mL/dL) represents dissolved O_2. Note also that the difference between the arterial and venous O_2 content ($CaO_2 - CvO_2$) is 5 mL/dL, or 50 mL/L, which means that 50 mL of O_2 is extracted from each liter of blood flowing through the capillaries. At a normal cardiac output of 5 L/min, the O_2 extracted from capillary blood would be $5 \times 50 = 250$ mL/min, which is the normal O_2 consumption in an adult at rest. This demonstrates how *the oxygenation of blood can provide information about the oxygen utilization in tissues.*

Simplified O_2 Content Equation

The dissolved O_2 is such a small fraction of the total O_2 content that it is usually eliminated from the equation describing O_2 content, as shown below.

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$$O_2 \text{ Content} = 1.34 \times [Hb] \times SO_2 \quad (9.6)$$

Volume Distribution of O_2 in Blood

Although the O_2 content in arterial blood exceeds that in venous blood, the volume of O_2 (i.e., the product of the O_2 content and blood volume), in arterial blood is far lower than in venous blood. This is shown in Table 9.1, where the volume of O_2 in arterial blood (250 mL) is less than half the volume in venous blood (555 mL). This is a reflection of the uneven distribution of blood volume in arteries and veins: i.e., the arteries contain only 25% of the total blood volume, while the remaining 75% is in the venous circulation. Thus, of the total volume of O_2 in the blood (250 + 555 = 805 mL), about 70% (555/805) is not available for aerobic metabolism because it is in the venous circulation.

OXYGEN TRANSPORT

The transport of O_2 from the lungs to metabolizing tissues is described by three parameters: the rate of O_2 delivery in arterial blood (O_2 delivery), the rate of O_2 uptake from the systemic microcirculation (O_2 uptake), and the fractional extraction of O_2 from capillary blood (O_2 extraction ratio). The reference ranges for these measures are listed in Table 9.2, including both absolute and size-adjusted ranges. The preferred measure of body size for hemodynamic parameters is the body surface area (BSA) in square meters (m^2). (See Equation 8.3 for the calculation of BSA.) An average-sized adult has a BSA of about 1.7 m^2.

TABLE 9.2 Normal Ranges for Oxygen Transport Parameters

Parameter	Absolute Range	Size-Adjusted Range†
Cardiac Output	5–6 L/min	2.4–4.0 L/min/m^2
O_2 Delivery	900–1,100 mL/min	520–600 mL/min/m^2
O_2 Uptake	200–270 mL/min	110–160 mL/min/m^2
O_2 Extraction Ratio	0.2–0.3	

†Size-adjusted values are the absolute values divided by the patient's body surface area in square meters (m^2).

Oxygen Delivery (DO_2)

The rate of O_2 delivery in arterial blood (DO_2) is the product of the cardiac output (CO) and the O_2 content of arterial blood (CaO_2) (5); i.e.,

$$DO_2 = CO \times CaO_2 \times 10 \text{ (mL/min)} \quad (9.7)$$

(The multiplier of 10 is used to convert the CaO_2 from mL/dL to mL/L.) If the CaO_2 is broken down into its components (1.34 × [Hb] × SaO_2), Equation 9.7 can be rewritten as:

$$DO_2 = CO \times (1.34 \times [Hb] \times SaO_2) \times 10 \quad (9.8)$$

Three measurements are needed to calculate the DO_2: cardiac output, hemoglobin concentration, and arterial O_2 saturation. If the DO_2 is expressed in relation to body size (BSA), the "cardiac index" (CI = CO/BSA) is used instead of the cardiac output. The DO_2 in healthy adults at rest is 900–1,100 mL/min, or 500–600 mL/min/m^2 when ad-justed for body size.

Oxygen Uptake

The rate of O_2 transport from the systemic capillaries into the tissues is called the *oxygen uptake* (VO_2). Since oxygen is not stored in tissues, the VO_2 is also a measure of the *oxygen consumption* in systemic tissues. The VO_2 can be calculated as the product of the cardiac output (CO) and the difference between arterial and venous O_2 content ($CaO_2 - CvO_2$).

$$VO_2 = CO \times (CaO_2 - CvO_2) \times 10 \text{ (mL/min)} \quad (9.9)$$

(The multiplier of 10 is included for the same reason as explained for the DO_2.) This equation is a modified version of the Fick equation for cardiac output (i.e., $CO = VO_2 / CaO_2 - CvO_2$), and has been called the *reverse Fick method* (6). The CaO_2 and CvO_2 in Equation 9.9 share a common term (1.34 × [Hb]), so the equation can be restated as:

$$VO_2 = CO \times 1.34 \times [Hb] \times (SaO_2 - SvO_2) \times 10 \quad (9.10)$$

Thus, four measurements are required to calculate the VO_2: the same 3 measurements used to calculate the DO_2, plus the O_2 saturation in "mixed" venous blood in the pulmonary arteries (SvO_2), which requires a pulmonary artery catheter. When the VO_2 is expressed in relation to body size (BSA), the "cardiac index" (CI = CO/BSA) is used instead of the cardiac output. The VO_2 in healthy adults at rest is 200–300 mL/min, or 110–160 mL/min/m^2.

Variability

Each of the 4 measurements used to derive the VO_2 has an inherent variability, and these are shown in Table 9.3 (6–8). The variability of the calculated VO_2 is ±18%, which is the summed variability of the individual components. Therefore, *the VO_2 that is calculated from the modified Fick equation must change by at least 18% for the change to be considered significant.*

Calculated vs. Measured VO_2

The calculated VO_2 is not the whole-body VO_2 because it does not include the VO_2 of the lungs. Normally, the VO_2 of the lungs accounts for less than 5% of the whole-body VO_2 (9), but it can make up 20% of the whole body VO_2 when there is inflammation in the lungs (10).

MEASURED VO_2: The measurement of the whole body VO_2 involves simultaneous measurements of the O_2 concentration in inhaled and exhaled gas. This requires an instrument that is equipped with an oxygen analyzer, such as the "metabolic carts" used for indirect calorimetry (see Chapter

48). The O_2 analyzer is connected to the proximal airway (usually in intubated patients) to record the fractional concentration of O_2 in inhaled and exhaled gas (F_IO_2 and F_EO_2). The instrument also records the minute ventilation (V_E), and the VO_2 is derived as follows:

$$VO_2 = VE \times (F_IO_2 - F_EO_2) \quad (9.11)$$

The measured VO_2 has a reported variability of ±5% (6,8), which is much less than the variability of the calculated VO_2 (±18%). The major drawback of the VO_2 measurement is the need for specialized equipment and trained personnel, which limits availability.

TABLE 9.3 Variability of Measurements Used to Calculate the Oxygen Consumption

Measurement	Variability
Thermodilution Cardiac Output	± 10%
Hemoglobin Concentration	± 2%
O_2 Saturation of Hemoglobin	± 2%
O_2 Content of Blood	± 4%
CaO_2–CvO_2	± 8%
Calculated VO_2	± 18%
Measured VO_2	± 5%

From References 6–8.

Oxygen Extraction Ratio

The fractional uptake of O_2 into tissues is expressed as the *oxygen extraction ratio* (O_2ER), which is the ratio of O_2 uptake (VO_2) to O_2 delivery (DO_2).

$$O_2ER = VO_2 / DO_2 \quad (9.12)$$

This ratio can be multiplied by 100 and expressed as a percentage. The VO_2 and DO_2 share common terms (Q × 1.34 x [Hb] x 10), which allows Equation 9.12 to be restated as follows:

$$O_2ER = (SaO_2 - SvO_2) / SaO_2 \quad (9.13)$$

If the SaO_2 is close to 100% (close to 1.0), the denominator in Equation 9.13 can be eliminated; i.e.,

$$O_2ER = (SaO_2 - SvO_2) \quad (9.14)$$

or the equation can be futher simplified to include only a single variable:

$$O_2ER = 1 - SvO_2 \quad (9.15)$$

The VO_2 is normally about 25% of the DO_2, so the normal O_2ER is 0.25 (range = 0.2–0.3), as shown in Table 9.2. Thus, *only 25% of the O_2 delivered to the capillaries is taken up into the tissues* when conditions are normal. This changes when O_2 delivery is reduced, as described next.

OXYGEN EXTRACTION

The oxygen transport system operates to maintain a constant VO_2 in the face of decreases in O_2 delivery (DO_2), and this is accomplished by compensatory increases in the O_2 extraction ratio (O_2ER) (11). This is explained by rearranging the terms in Equation 9.12 so that VO_2 is the derived variable:

$$VO_2 = DO_2 \times O_2ER \tag{9.16}$$

This equation predicts that the VO_2 will remain constant when DO_2 is decreased if there is an proportional increase in O_2 extraction. However, if the increase in O_2 extraction cannot fully compensate for a decrease in DO_2, then the VO_2 will begin to decrease, and this marks the onset of oxygen-limited (anaerobic) metabolism.

The relationships in Equation 9.16 are illustrated in Figure 9.3. The graph shows the effects of a progressive decrease in DO_2 on the VO_2, and the boxes show the O_2 extraction at relevant points on the curve. In this case, O_2 extraction is represented by the (SaO_2–SvO_2) difference, which is allowed because the SaO_2 is typically maintained at >90% (i.e., close to 1.0). This creates the following relationships:

$$VO_2 = DO_2 \times (SaO_2 - SvO_2) \tag{9.17}$$

The graph indicates that, as DO_2 decreases below normal (moving to the left along the curve), the VO_2 initially remains unchanged, indicating that there is a compensatory increase in (SaO_2–SvO_2). However, a point is eventually reached where the VO_2 begins to decrease, and at this point, the (SaO_2–SvO_2) difference has increased from 25% to 50%. Beyond this point, the VO_2 decreases in response to a decrease in DO_2, indicating that O_2 extraction is no longer able to fully compensate for the declining DO_2. The point where the VO_2 becomes “delivery-dependent” is the threshold for O_2-limited metabolism (12).

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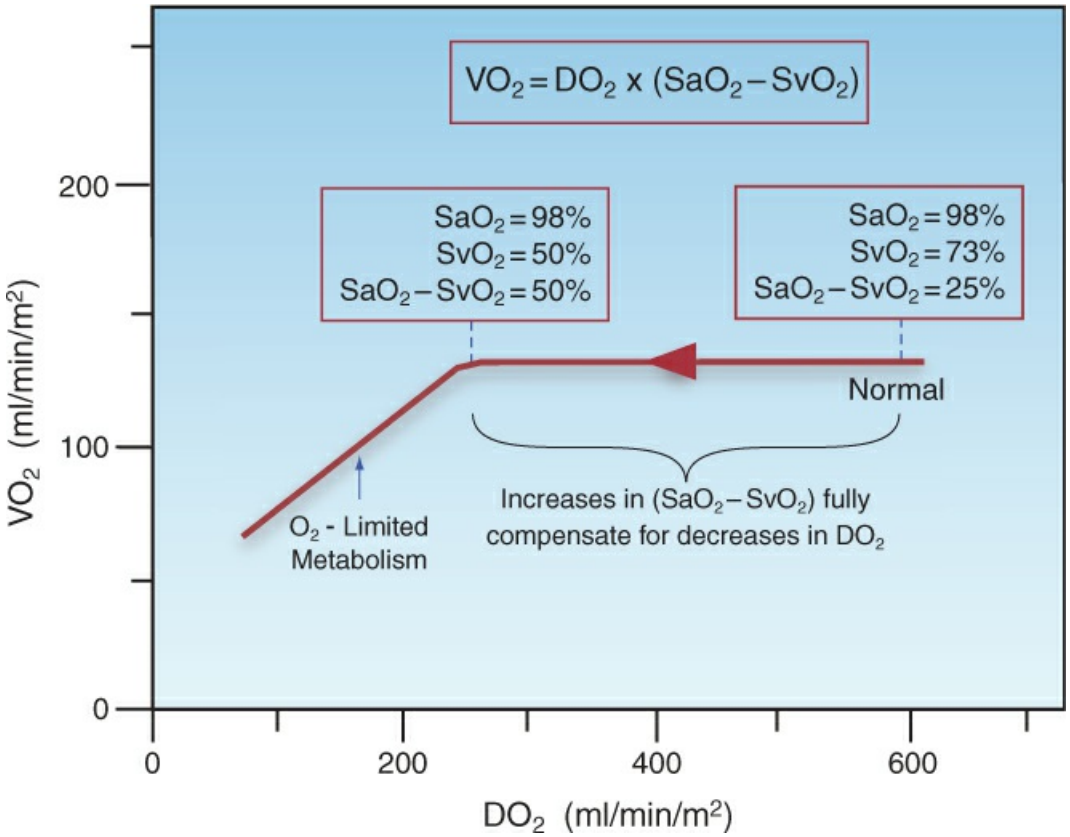


FIGURE 9.3 Graph showing the relationship between O_2 delivery (DO_2) and O_2 uptake (VO_2). The O_2 extraction ratio (O_2ER) is represented by (SaO_2 – SvO_2) difference. See text for explanation).

The relationships in Figure 9.3 indicate that the (SaO_2–SvO_2) difference can be used to assess the balance between O_2 delivery and O_2 consumption, and more importantly, to identify when tissue oxygenation is threatened or impaired.

The (SaO_2 – SvO_2) Difference

As shown in Equation 9.14, the (SaO_2–SvO_2) difference can be used as a measure of O_2 extraction as long as the SaO_2 exceeds 90%. The following are some guidelines for the interpretation of the (SaO_2–SvO_2) difference, which are summarized in Table 9.4. (*Note:* O_2 extraction is influenced by the metabolic rate, and the following guidelines are valid only if the metabolic rate is normal.)

. The normal (SaO_2– SvO_2) is 20–30%.

. An increase in (SaO_2– SvO_2) above 30% indicates a decrease in O_2 delivery (from hypoxemia, anemia, or decreased cardiac output).

. An increase in (SaO_2– SvO_2) to ≥50% indicates that tissue oxygenation is threatened or impaired. This has been validated in animal studies (13).

. A decrease in (SaO_2– SvO_2) to <20% indicates a defect in O_2 utilization in tissues, which is

usually the result of sepsis. This condition has also been called “peripheral shunting”, since it mimics the changes produced by peripheral arteriovenous shunts.

The SaO_2 is monitored continuously with pulse oximetry (which is mandatory in ICU patients), and the SvO_2 is measured in “mixed venous” blood in the pulmonary arteries, which requires a pulmonary artery (PA) catheter. The SvO_2 can be monitored continuously with specialized PA catheters capable of venous oximetry (e.g., Triox™-PAC, ICU Medical, San Clemente, CA). The technique of venous oximetry is described in Chapter 7, and illustrated in Figure 7.5.

Venous O_2 Saturation (SvO_2)

When the SaO_2 approaches 100%, the SvO_2 can be used alone to evaluate O_2 extraction, as described in Equation 9.15. This equation demonstrates that the SvO_2 varies inversely with changes in O_2 extraction: e.g., an increase in O_2 extraction will result in a decrease in SvO_2. The following are some guidelines for interpreting the SvO_2, which are summarized in Table 9.4.

. The normal range for SvO_2 in pulmonary artery blood is 65–75% (14).

. The SvO_2 varies spontaneously by an average of 5% (15), so a change in SvO_2 should exceed 5% to be considered significant.

. A decrease in SvO_2 to <65% indicates a decrease in O_2 delivery (from hypoxemia, anemia, or decreased cardiac output).

. A decrease in SvO_2 to ≤50% indicates that tissue oxygenation is threatened or impaired (16).

. An SvO_2 that is 80% or higher is a sign of impaired O_2 utilization in metabolizing tissues, and is usually the result of sepsis.

TABLE 9.4 Surrogate Measures of O_2 Extraction

Measure	Normal Range	Tissue O_2 Threatened	Defect in O_2 Utilization
$SaO_2 - SvO_2$	20-30%	≥50%	<20%
SvO_2	65-75%	≤50%	>80%

Central Venous O_2 Saturation ($ScvO_2$)

The decline in popularity of pulmonary artery catheters has shifted attention to the O_2 saturation in the superior vena cava; i.e., the “central venous O_2 saturation” ($ScvO_2$). The relevant information about $ScvO_2$ is summarized in the following statements.

. The $ScvO_2$ is lower than the SvO_2 in healthy subjects. However, it is higher than the SvO_2 (by an average of 7%) in critically ill patients (14,17), and this difference can be as high as 18% in patients with circulatory shock (17,18). The $ScvO_2$ is heavily influenced by the cerebral

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venous O_2 saturation, and the elevated $ScvO_2$ in circulatory shock is attributed to peripheral vasoconstriction with relative preservation of cerebral blood flow.

. Despite the discrepancy between $ScvO_2$ and SvO_2, changes in $ScvO_2$ generally mirror those in the SvO_2 (17), and *trends in the $ScvO_2$ are considered more reliable than individual measurements* (19). The definition of a significant change in $ScvO_2$ is the same as that mentioned for the SVO_2 (i.e., Δ >5%).

. The normal range of $ScvO_2$ is not well defined, but an $ScvO_2$ of 70% is one of the recommended goals of hemodynamic management in circulatory shock (20).

The $ScvO_2$ can be monitored with central venous catheters or peripherally-inserted central catheters (PICCs); both catheters are available as oximetry catheters (e.g., Triox™-CVC and Triox™-PICC, ICU Medical, San Clemente, CA) that provide continuous $ScvO_2$ monitoring.

Tissue O_2 Saturation (StO_2)

The technique of *near-infrared spectroscopy* (NIRS) is a form of oximetry that uses an optical probe on the skin to measure the O_2 saturation of hemoglobin in the microcirculation of the underlying tissue. This measurement is known as the *tissue O_2 saturation* (StO_2), which is a misleading term because the measured variable is mostly in blood. Since most of the blood in the microcirculation is in the venules, the StO_2 is primarily a reflection of venous O_2 saturation (SvO_2 or $ScvO_2$) (21). (The StO_2 in muscle can also be influenced by the O_2 saturation of myoglobin) (22).

In adults, NIRS has been used to monitor the StO_2 in the cerebral cortex (23) and in skeletal muscle (24). In patients with circulatory shock, there is a close correlation between changes in skeletal muscle StO_2 and changes in systemic O_2 delivery (DO_2), suggesting that StO_2 monitoring could serve as a noninvasive means of tracking changes in O_2 delivery (24).

One of the major limitations of the StO_2 measurement is the inability to define normal and abnormal levels, since there is a marked variability in the StO_2 in healthy subjects, and in patients with circulatory shock (25). This could be a reflection of the methodology, because photons scatter in tissues (i.e., they do not follow a straight path from light source back to the photodetector), and the degree of scatter can vary greatly in individual subjects and in different pathological states.

LACTATE

An increase in the plasma lactate concentration (>2 mmol/L) is considered the most reliable marker of inadequate tissue oxygenation, and it is a stated requirement for the diagnosis of circulatory shock (26). However, the traditional concepts about lactate are changing, as explained in this section.

Lactate and Survival

Clinical studies have consistently shown that the mortality rate in critically ill patients is directly

related to the initial plasma lactate level, and to the time required for the lactate level to return to normal after treatment is initiated. This is demonstrated in Figure 9.4. The graph on the left shows the relationship between the initial lactate level and in-hospital mortality (27), and the graph on the right shows the relationship between mortality and the time required to normalize the plasma lactate (called "lactate clearance") (28).

The relationship between plasma lactate levels and survival highlights the importance of monitoring plasma lactate in critically ill patients. However, this relationship is *not* evidence that lactate is a marker of inadequate tissue oxygenation, as explained next.

Circular Reasoning

The relationship between plasma lactate levels and survival is considered evidence that lactate is a marker of inadequate tissue oxygenation because of the popular notion that oxygen deprivation is the universal source of cell death in critically ill patients. However, this notion is largely *based* on the relationship between plasma lactate levels and survival.

Contrary Evidence

Because of the difficulties in recording the PO_2 in cells (and especially in mitochondria), there is no evidence that cell death in critically ill patients is the result of O_2 deprivation. However, there is evidence to the contrary: i.e., in animal studies using a positron-emitting marker of cell hypoxia (18F-fluoromisonidazole), there was no evidence of cell hypoxia in any of the major organs in septic shock (29), and human studies using PO_2 recordings from skeletal muscle have reported that *tissue PO_2 levels are actually increased in patients with septic shock* (30,31).

Considering that the intracellular PO_2 in skeletal muscle averages 5 mm Hg, and can be <1 mm Hg (32), while the recorded PO_2 in cardiac myocytes is 0.2–2.4 mm Hg (33), it seems that cells normally operate at very low PO_2 levels, which is contrary to the idea that cell death is the result of low PO_2 levels. In fact, an oxygen-poor environment in cells is advantageous because it limits the risk of oxidative cell injury.

The observations just presented suggest that cellular O_2 deprivation may not be the universal culprit in cell death (and organ failure) in critically ill patients, and this implies that aerobic lactate production may be much more common than suspected (see next).

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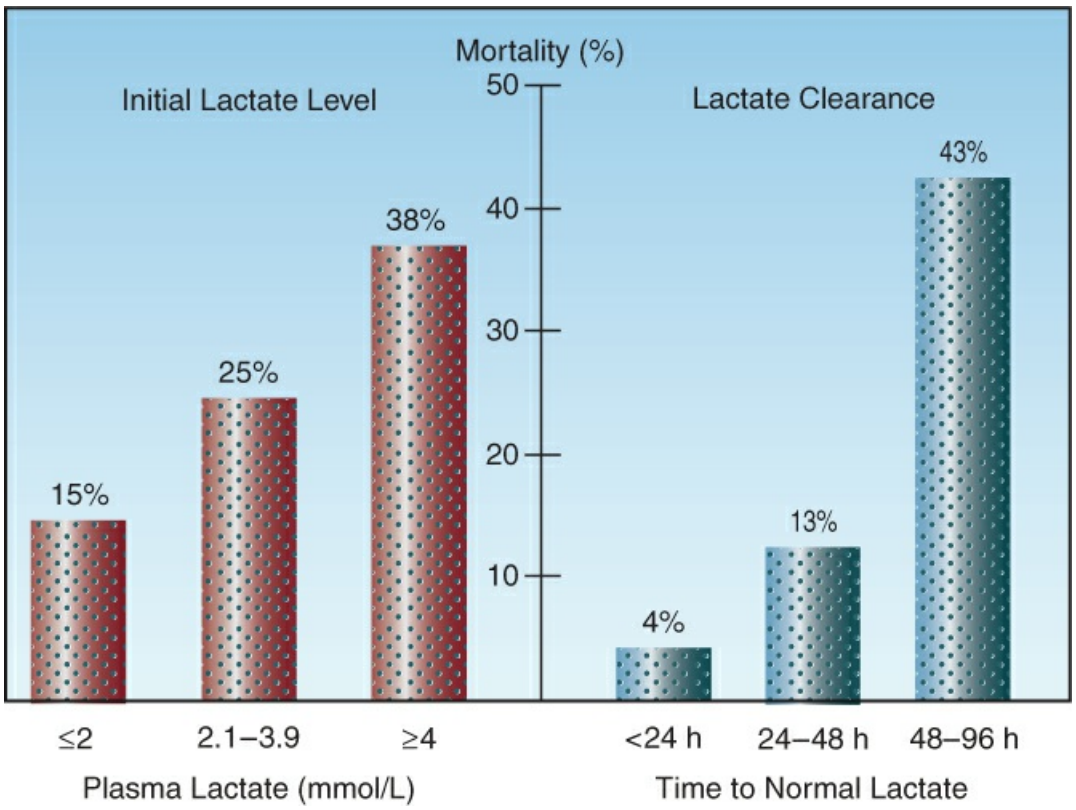


FIGURE 9.4 Graphs showing the relationship between plasma lactate levels and in-hospital mortality in critically ill patients. The graph on the left shows the relationship between the initial lactate level and mortality (27), and the graph on the right shows the relationship between the mortality rate and the time for increased lactate levels to normalize (lactate clearance) (28).

Aerobic Lactate Production

The role of aerobic lactate production is explained using the pathways of glucose metabolism in Figure 9.5. There are two general pathways for glucose metabolism: one is located in the cytoplasm, and is known as *glycolysis*, and the other is located in mitochondria, and is involved in *oxidative phosphorylation* (i.e., the production of ATP). The glycolytic pathway operates in the presence or absence of oxygen, while the mitochondrial pathway proceeds only in the presence of oxygen. The traditional teaching is that glycolysis reaches a pivotal point with the formation of pyruvate: when O_2 is available, pyruvate moves into mitochondria via the enzyme pyruvate dehydrogenase, and when O_2 is not available, pyruvate is reduced to lactate using the enzyme lactate dehydrogenase (LDH).

According to the scheme just presented, pyruvate is the end-point of aerobic glycolysis, and lactate is the end-point of anaerobic glycolysis. However this is not the case, because lactate is produced under aerobic conditions: i.e., the daily production of lactate averages about 20 mmol/kg body weight (34). In fact, the reaction that converts pyruvate to lactate has an equilibrium constant that strongly favors lactate formation. The aerobic production of lactate is enhanced in several clinical conditions, including those listed in Table 9.5. Many of the conditions in this table are likely to be encountered in critically ill patients, including those described next.

TABLE 9.5 Sources of Aerobic Hyperlactatemia	
Conditions	**Drugs and Toxins**
Asthma (severe)	β Agonists
Ketoacidosis	Catecholamines
Liver Dysfunction	Cyanide
Seizures	Metformin
Sepsis	Propofol
Stress	Propylene Glycol
Thiamine Deficiency	Salicylates
Tumors	Toxic Alcohols

Sepsis

Hyperlactatemia is considered a universal consequence of sepsis and septic shock (26, 35), and two sources of this phenomenon have been identified. The principal source is inhibition of the pyruvate dehydrogenase (PDH) enzyme that allows pyruvate to enter mitochondria and fuel oxidative phosphorylation (see Figure 9.5) (36). The culprits in this inhibition are bacterial products (e.g., endotoxin) and proinflammatory cytokines (37,38). The result is enhanced lactate production that is not the result of cellular hypoxia. Sepsis-induced inhibition of the PDH enzyme would explain why the cell injury in sepsis is attributed to a defect in O_2 utilization in mitochondria (38). Sepsis is also associated with a catecholamine-driven increase in the rate of glycolysis (39), and this (especially when combined with inhibition of the PDH enzyme) is the second source of aerobic lactate production in sepsis.

As just described, there is considerable evidence that the hyperlactatemia in sepsis is the result of aerobic (not anaerobic) lactate production. Considering that sepsis is the leading cause of in-hospital deaths in the United States (40), it is reasonable to conclude that aerobic hyperlactatemia is much more common than suspected in critically ill patients.

Physiological Stress

Stressful conditions are associated with an increase in glucose uptake into cells, which is attributed to the accumulation of a glucose transporter protein at the cell surface (41). This effect, combined with the stimulatory effect of catecholamines mentioned previously (39), results in a considerable (40–50%) increase in the rate of aerobic glycolysis, and a subsequent increase in aerobic lactate production (29).

Thiamine Deficiency

The PDH enzyme that draws pyruvate into the mitochondrion (and thereby fuels oxidative phosphorylation) requires thiamine pyrophosphate (TPP) as a cofactor (see Figure 9.5), and deficiency of this cofactor will divert pyruvate to the production of lactate and enhance aerobic lactate production. This is the mechanism for the hyperlactatemia in thiamine deficiency (42), which is considered a common and often unrecognized condition in critically ill patients (43).

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(See Chapter 48 for more on thiamine deficiency.)

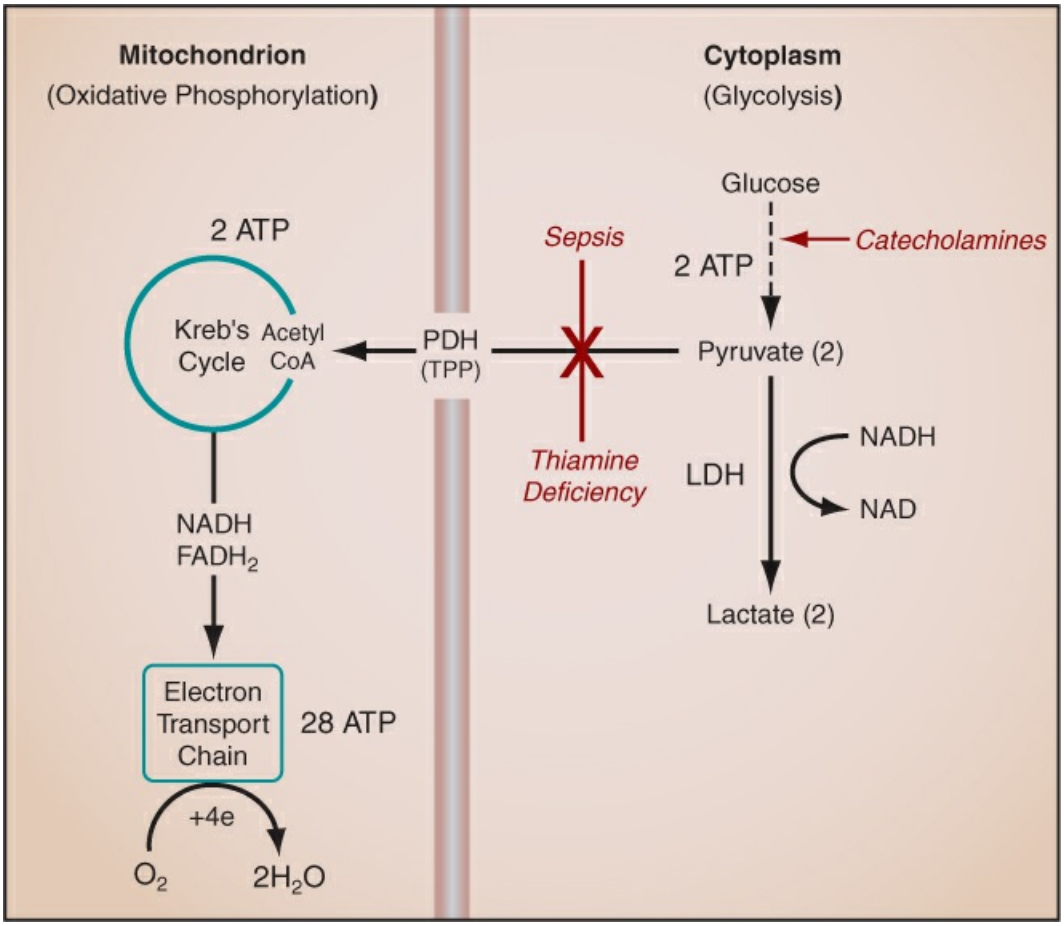


FIGURE 9.5 The pathways of glucose metabolism. ATP = adenosine triphosphate, PDH = pyruvate dehydrogenase, TPP = thiamine pyrophosphate, LDH = lactate dehydrogenase, NAD and NADH = nicotinamide adenine dinucleotide (oxidized and reduced), $FADH_2$ = flavin adenine dinucleotide (reduced). See text for further explanation.

Conclusion

The information just presented indicates that *increased lactate production in critically ill patients may be predominantly aerobic in origin*. In fact, anaerobic lactate production is likely to be an uncommon event, as stated in a recent review of this topic (44):

> Today, it is well established that any increase in lactate concentration typically represents something other than O_2 limitation; hypoxia-driven lactate accumulation is very much the exception rather than the rule.

So why is lactate production increased in critical illness? One possible answer is presented next.

Lactate as an Oxidative Fuel

Lactate can be viewed, not as a waste product of anaerobic metabolism, but rather as an alternative fuel source during periods of metabolic stress (39,44). The energy yield from the oxidative metabolism of lactate is equivalent to that of glucose, and this is shown in Table 9.6. The caloric density (kcal/g) of lactate and glucose are equivalent and, since one glucose molecule

produces 2 lactate molecules, the energy yield from the complete oxidation of lactate and glucose are equivalent. Thus, lactate can serve as a source of energy when glucose availability is threatened by heightened metabolic demands.

TABLE 9.6 Glucose vs. Lactate as Oxidative Fuels

Substrate	Molecular Weight (g/mole)	Energy Yield (kcal/mole)	Energy Yield (kcal/g)
Glucose	180	673	3.74
Lactate	90	326	3.62
Lactate x 2	180	652	3.62

In conditions of metabolic stress (such as sepsis), lactate is often used as an oxidative fuel by the heart and brain. In this situation, lactate can provide 60% of the energy needs of the myocardium (44), and there is evidence that lactate improves cardiac performance in circulatory shock (45). Lactate can also provide about 25–30% of the energy needs of the brain in stressful conditions (46). Adaptations like these help to preserve the viability of the heart and brain in life-threatening conditions.

A FINAL WORD

Is Cellular Hypoxia a Common Cause of Death?

Since it is not possible to monitor tissue O_2 levels, the notion that impaired tissue oxygenation is responsible for multiorgan failure and fatal outcomes in critically ill patients is largely based on studies showing a correlation between plasma lactate levels and mortality. However, this interpretation is based on the assumption that lactate elevation is a marker of anaerobic metabolism in critically ill patients, which is likely not the case (as presented in the latter part of this chapter).

There is evidence (from animal studies) that intracellular PO_2 levels are as low as 1 mm Hg (47), and further, that aerobic production of ATP can continue down to PO_2 levels of ≤0.5 mm Hg (33,48). These observations indicate that *there is an oxygen-poor environment in the interior of cells, and that aerobic metabolism is designed to operate in such an environment* (49). As such, cell injury from a low PO_2 seems unlikely. The culprit in multiorgan failure is more likely to be inflammatory cell injury, especially in sepsis/septic shock, which is the leading cause of in-hospital deaths in the United States (40).

So, why is there an oxygen-poor environment in our cells? For the same reason that we store food in vacuum-sealed containers, and wrap our sandwiches in cellophane; i.e., because *oxygen disrupts or decomposes organic (carbon-based) matter*, which includes all of our vital cell components. (See Chapter 25 for more on oxygen-related cell injury.)

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